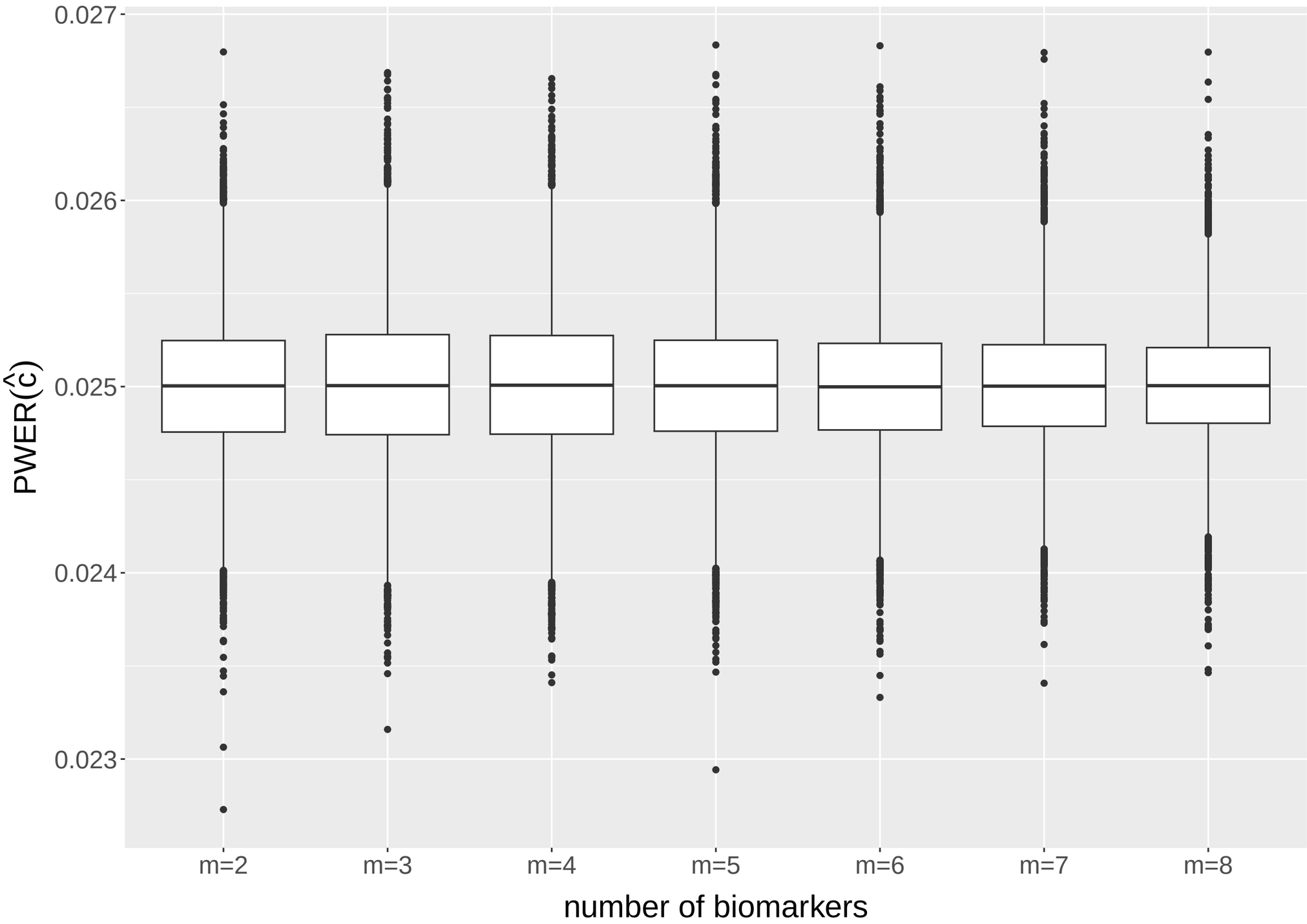
Distribution of the true PWER

Figure	Prevalence estimate				Test statistics	Allocation in strata	sample size	number of biomarkers	Mean	SD	Min	Q1	Med	Q3	Max
1	MLE	independent biomarkers	random prevalences	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 2$	0.025	0.00039	0.02273	0.02476	0.025	0.02525	0.0268
								$m = 3$	0.02501	0.00042	0.02316	0.02474	0.02501	0.02528	0.02669
								$m = 4$	0.02501	0.00041	0.02341	0.02474	0.02501	0.02527	0.02665
								$m = 5$	0.02501	0.00039	0.02294	0.02476	0.025	0.02525	0.02683
								$m = 6$	0.025	0.00037	0.02333	0.02477	0.025	0.02523	0.02683
								$m = 7$	0.02501	0.00035	0.02341	0.02479	0.025	0.02523	0.02679
								$m = 8$	0.02501	0.00033	0.02346	0.0248	0.02501	0.02521	0.0268
2	MLE	correlated biomarkers	random prevalences	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 2$	0.02502	0.00039	0.02226	0.02477	0.02502	0.02526	0.02656
								$m = 3$	0.025	0.00042	0.02286	0.02473	0.025	0.02527	0.02663
								$m = 4$	0.02502	0.00041	0.0232	0.02476	0.02501	0.02527	0.02689
								$m = 5$	0.02501	0.00039	0.02317	0.02475	0.025	0.02525	0.02696
								$m = 6$	0.02501	0.00037	0.02353	0.02477	0.02501	0.02524	0.02678
								$m = 7$	0.02501	0.00035	0.02357	0.02479	0.02501	0.02522	0.02671
								$m = 8$	0.02501	0.00032	0.02365	0.0248	0.025	0.02521	0.02644
3	MLE		fixed prevalences (equal in all strata)	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 2$	0.02501	0.00038	0.02357	0.02474	0.02499	0.02528	0.02632
								$m = 3$	0.02501	0.00043	0.02346	0.02472	0.025	0.0253	0.02672
								$m = 4$	0.025	0.00044	0.02334	0.0247	0.025	0.0253	0.02679
								$m = 5$	0.02501	0.00044	0.02355	0.02471	0.025	0.0253	0.02682
								$m = 6$	0.02501	0.00041	0.02369	0.02473	0.025	0.02529	0.02685
								$m = 7$	0.02501	0.00039	0.02363	0.02474	0.025	0.02527	0.02659
								$m = 8$	0.02501	0.00037	0.02347	0.02476	0.02501	0.02525	0.02663
4	MLE		fixed prevalences (0.5 in one stratum and equal in the others)	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 2$	0.02501	0.00037	0.02353	0.02477	0.025	0.02527	0.02648
								$m = 3$	0.02501	0.00049	0.02324	0.02468	0.02501	0.02534	0.0272
								$m = 4$	0.02502	0.00057	0.02279	0.02463	0.02501	0.0254	0.02731
								$m = 5$	0.02502	0.00063	0.02249	0.0246	0.02502	0.02545	0.02813
								$m = 6$	0.02503	0.00069	0.02255	0.02456	0.02501	0.02549	0.02814
								$m = 7$	0.02503	0.00073	0.02253	0.02453	0.02501	0.02551	0.02803
								$m = 8$	0.02503	0.00077	0.02245	0.02451	0.02502	0.02554	0.02846
5	marginal prevalence estimator	independent biomarkers	random prevalences	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 2$	0.02501	0.00027	0.02383	0.02484	0.02501	0.02517	0.02626
								$m = 3$	0.02501	0.00034	0.02381	0.02479	0.02501	0.02523	0.02638
								$m = 4$	0.02501	0.00036	0.02342	0.02477	0.02501	0.02524	0.02653
								$m = 5$	0.02501	0.00036	0.02335	0.02478	0.025	0.02524	0.0264
								$m = 6$	0.025	0.00035	0.02355	0.02478	0.025	0.02522	0.02651
								$m = 7$	0.02501	0.00034	0.02369	0.02479	0.025	0.02522	0.02651
								$m = 8$	0.025	0.00032	0.02351	0.0248	0.025	0.02521	0.02667
6	MLE	independent biomarkers	random prevalences	pairwise different treatments	z-distr.	equal	$N = 500$	$m = 2$	0.025	0.00039	0.02272	0.02476	0.025	0.02525	0.0268
								$m = 3$	0.02501	0.00042	0.02245	0.02474	0.02501	0.02528	0.02669
								$m = 4$	0.02501	0.00041	0.02341	0.02474	0.02501	0.02527	0.02666
								$m = 5$	0.02501	0.00039	0.02294	0.02476	0.025	0.02525	0.02684
								$m = 6$	0.025	0.00037	0.02333	0.02477	0.025	0.02523	0.02683
								$m = 7$	0.02501	0.00035	0.0234	0.02479	0.025	0.02523	0.0268
								$m = 8$	0.02501	0.00033	0.02345	0.0248	0.02501	0.02521	0.0268

Distribution of the true PWER

Figure	Prevalence estimate				Test statistics	Allocation in strata	sample size	number of biomarkers	Mean	SD	Min	Q1	Med	Q3	Max
7	MLE	independent biomarkers	random prevalences	one single treatment	t-distr.	equal	$N = 500$	$m = 2$	0.02501	0.00035	0.02282	0.02479	0.025	0.02522	0.02676
								$m = 3$	0.02502	0.00038	0.02332	0.02477	0.02501	0.02526	0.02663
								$m = 4$	0.02502	0.00038	0.02355	0.02478	0.02502	0.02526	0.02662
								$m = 5$	0.02502	0.00036	0.02309	0.02479	0.02502	0.02524	0.02682
								$m = 6$	0.02502	0.00035	0.02342	0.02479	0.02501	0.02524	0.02668
								$m = 7$	0.02503	0.00033	0.02347	0.02481	0.02502	0.02524	0.02674
								$m = 8$	0.02502	0.00031	0.02358	0.02483	0.02502	0.02522	0.02672
8	MLE	independent biomarkers	random prevalences	pairwise different treatments	t-distr.	random allocation probabilities	$N = 500$	$m = 2$	0.025	0.00039	0.02272	0.02476	0.025	0.02525	0.0268
								$m = 3$	0.02501	0.00042	0.02307	0.02474	0.02501	0.02528	0.02668
								$m = 4$	0.02501	0.00041	0.02339	0.02475	0.02501	0.02527	0.02665
								$m = 5$	0.02501	0.00039	0.02295	0.02476	0.025	0.02525	0.02684
								$m = 6$	0.025	0.00037	0.02334	0.02477	0.025	0.02523	0.02683
								$m = 7$	0.02501	0.00035	0.02341	0.02479	0.025	0.02523	0.0268
								$m = 8$	0.02501	0.00033	0.02345	0.0248	0.02501	0.02521	0.02681
9	MLE	independent biomarkers	random prevalences	pairwise different treatments	t-distr.	equal	$N = 25$	$m = 3$	0.02516	0.00183	0.01855	0.0239	0.02506	0.0263	0.03331
							$N = 50$		0.02508	0.00132	0.0201	0.02419	0.02504	0.02593	0.03043
							$N = 100$		0.02504	0.00094	0.02147	0.02441	0.02501	0.02563	0.02873
							$N = 150$		0.02503	0.00078	0.02113	0.02452	0.02501	0.02552	0.02879
							$N = 200$		0.02502	0.00067	0.02174	0.0246	0.02501	0.02544	0.02777
							$N = 500$		0.02501	0.00042	0.02316	0.02474	0.02501	0.02528	0.02669
10	MLE	independent biomarkers	random prevalences	pairwise different treatments	t-distr.	equal	$N = 25$	$m = 4$	0.02517	0.00176	0.01908	0.02401	0.02507	0.02619	0.03723
							$N = 50$		0.02507	0.00131	0.02024	0.02422	0.02503	0.02585	0.03219
							$N = 100$		0.02504	0.00093	0.02156	0.02445	0.02501	0.02562	0.02977
							$N = 150$		0.02503	0.00076	0.022	0.02454	0.025	0.02549	0.02845
							$N = 200$		0.02503	0.00066	0.02235	0.02459	0.02502	0.02543	0.02806
							$N = 500$		0.02501	0.00041	0.02341	0.02474	0.02501	0.02527	0.02665

Figure 1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0.026
0.025
0.024
0.023
PWER($\hat{c}$)
m=2
m=3
m=4
m=5
m=6
m=7
m=8
number of biomarkers

Figure 2

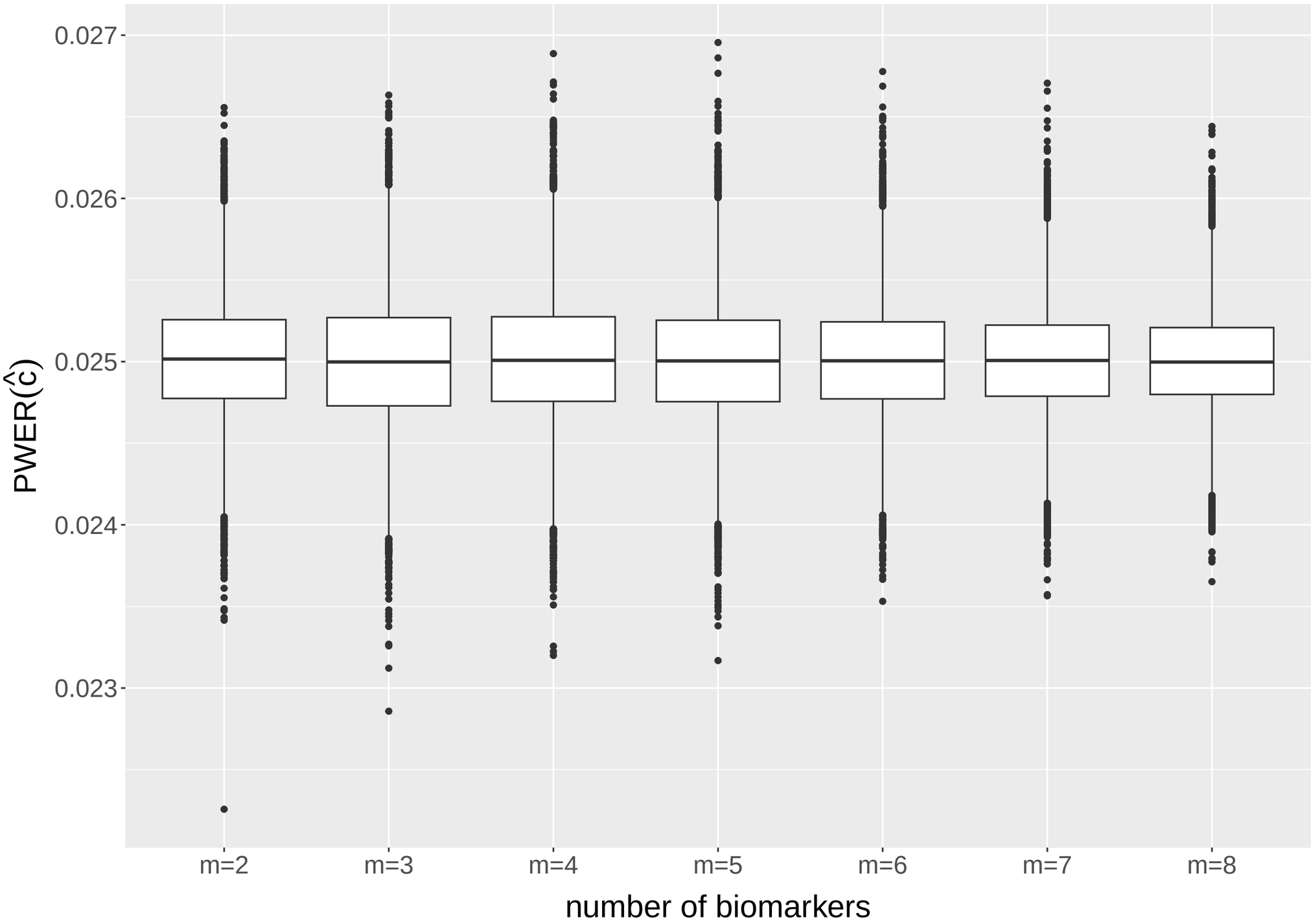

0.027
0.026
0.025
0.024
0.023
PWER($\hat{c}$)
m=2
m=3
m=4
m=5
m=6
m=7
m=8
number of biomarkers

Figure 3

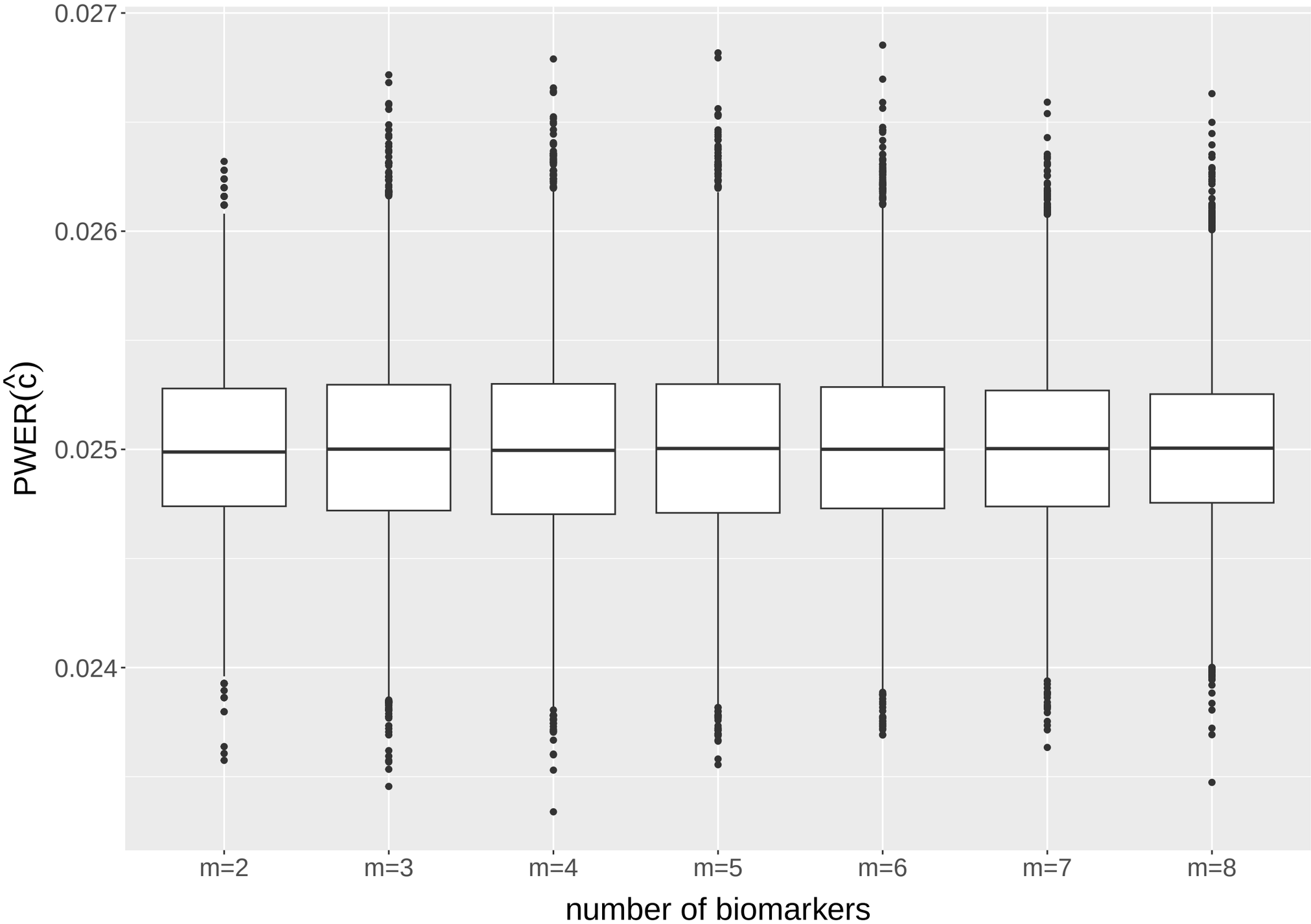

0.027
0.026
0.025
0.024
PWER($\hat{c}$)
m=2
m=3
m=4
m=5
m=6
m=7
m=8
number of biomarkers

Figure 4

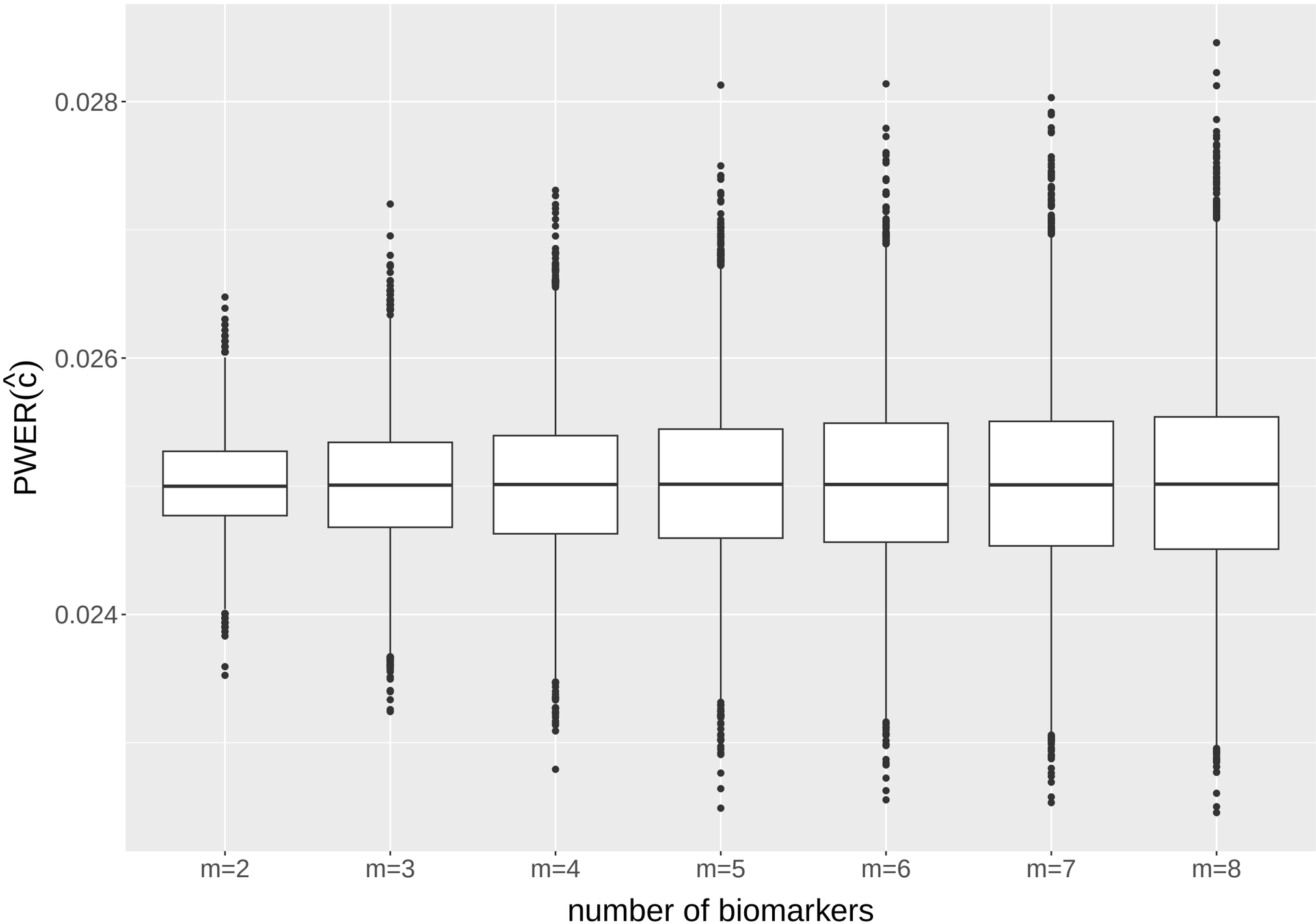

0.028
0.026
0.024
PWER($\hat{c}$)
m=2
m=3
m=4
m=5
m=6
m=7
m=8
number of biomarkers

Figure 5

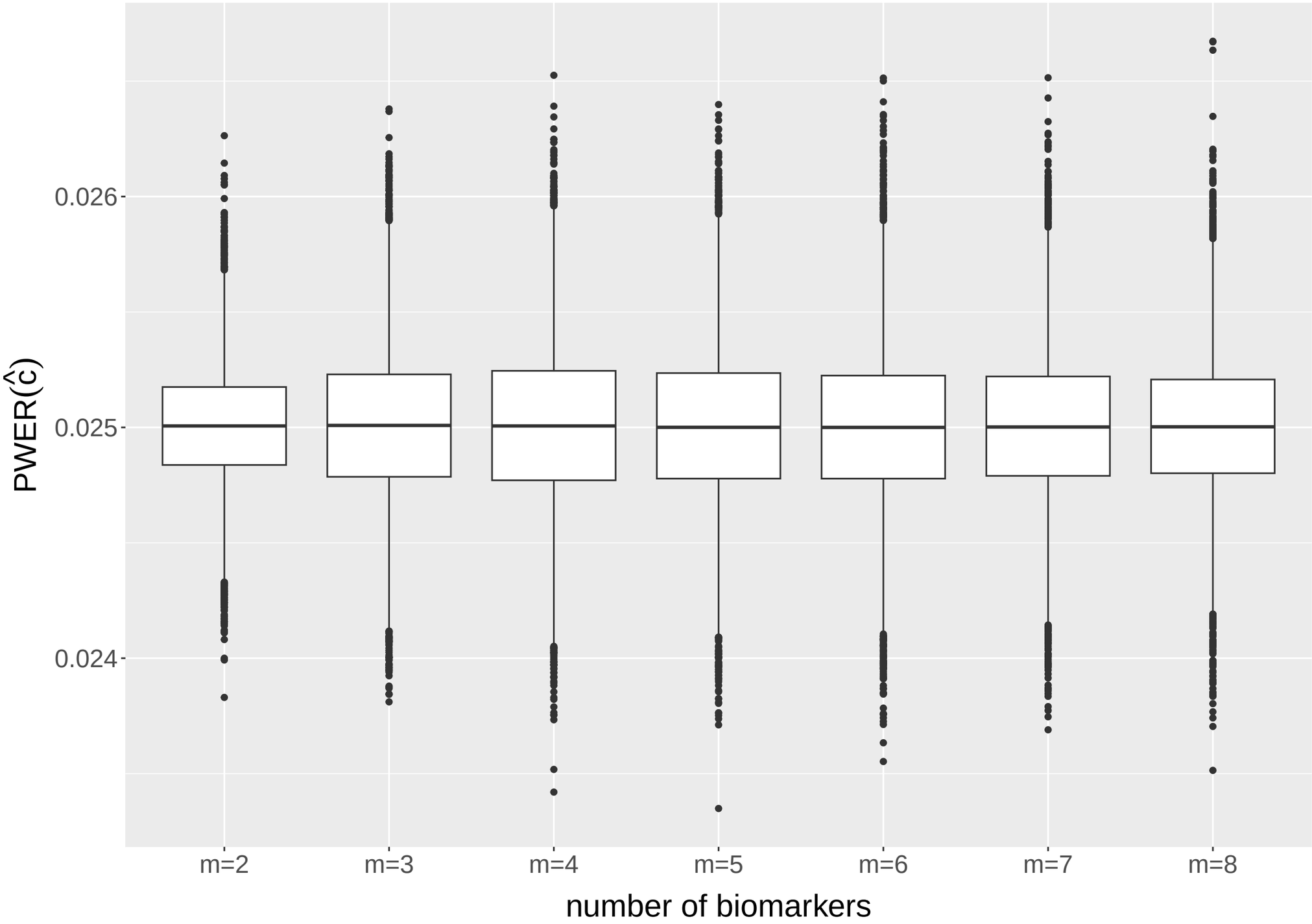

0.026
0.025
0.024
PWER($\hat{c}$)
m=2
m=3
m=4
m=5
m=6
m=7
m=8
number of biomarkers

Figure 6

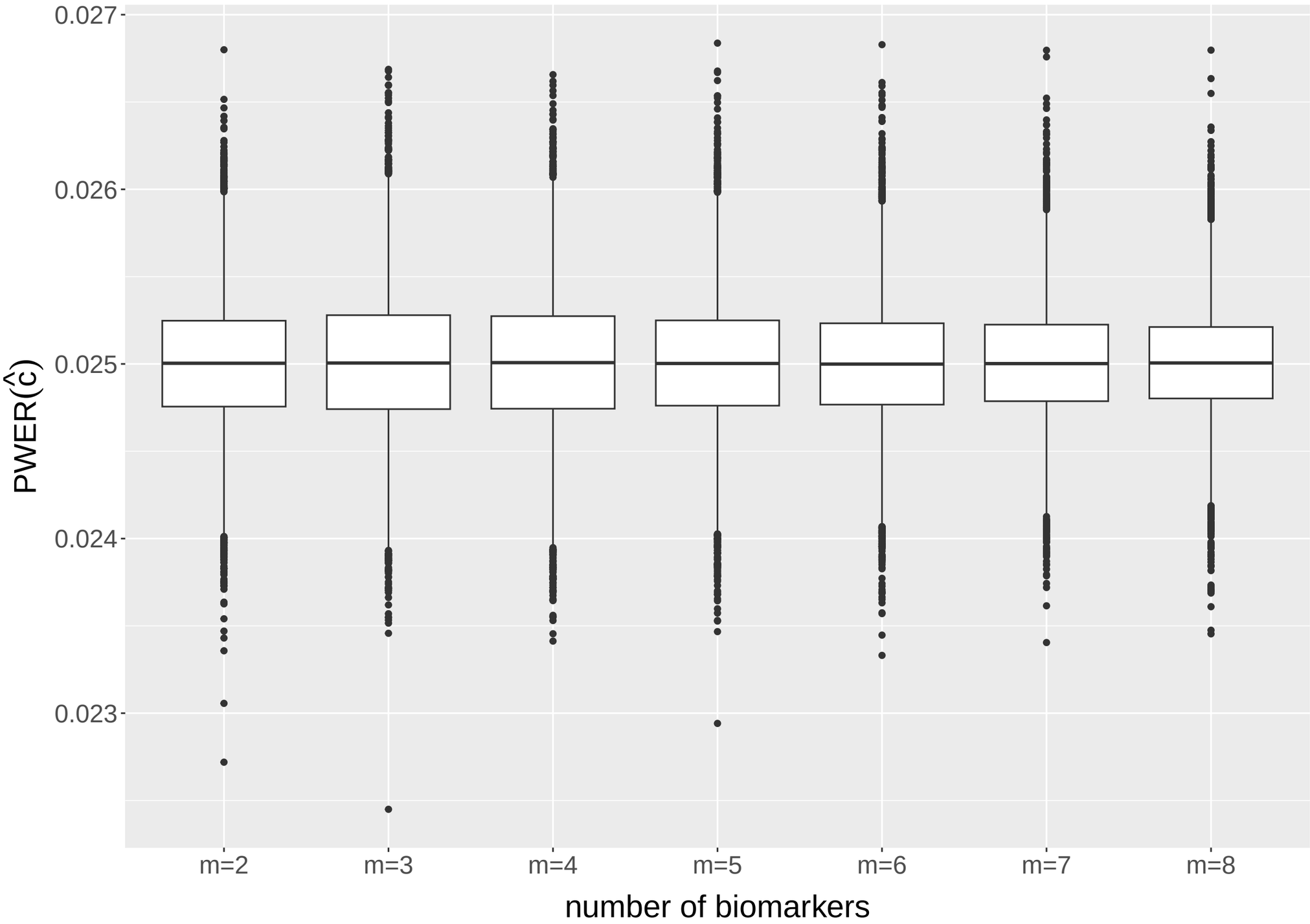

0.027
0.026
0.025
0.024
0.023
PWER($\hat{c}$)
m=2
m=3
m=4
m=5
m=6
m=7
m=8
number of biomarkers

Figure 7

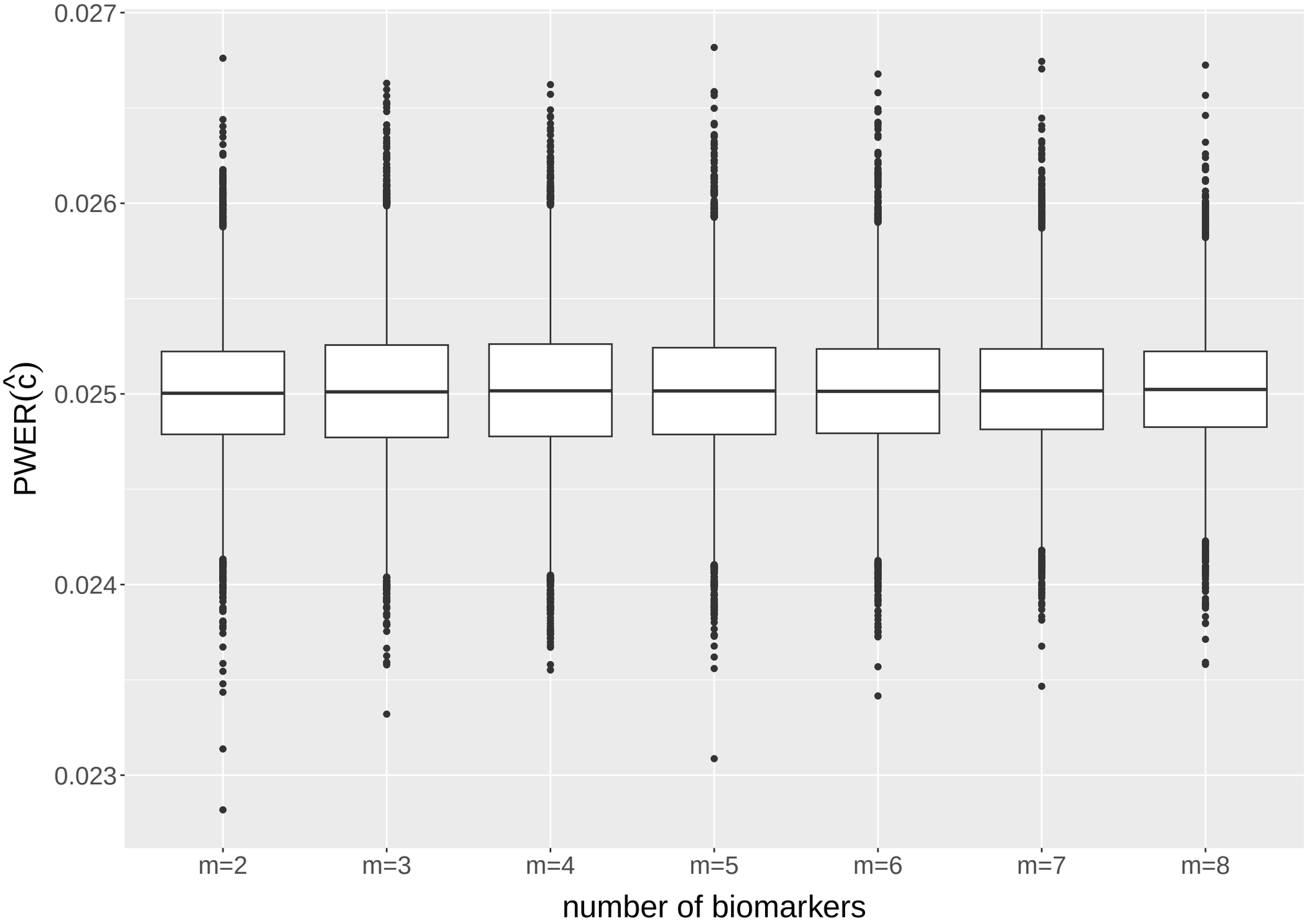

0.027
0.026
0.025
0.024
0.023
PWER($\hat{c}$)
m=2
m=3
m=4
m=5
m=6
m=7
m=8
number of biomarkers

Figure 8

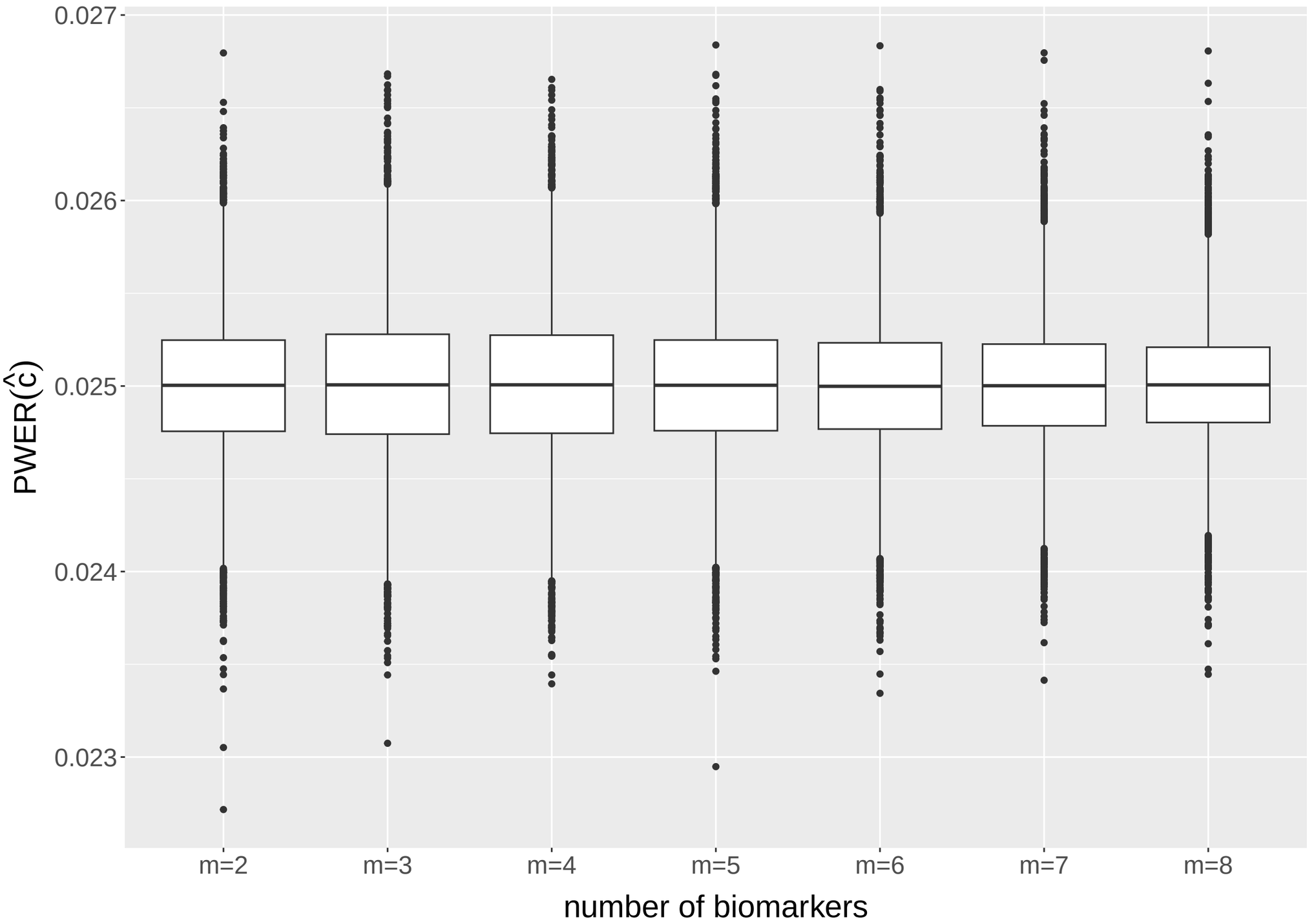

0.027
0.026
0.025
0.024
0.023
PWER($\hat{c}$)
m=2
m=3
m=4
m=5
m=6
m=7
m=8
number of biomarkers

Figure 9

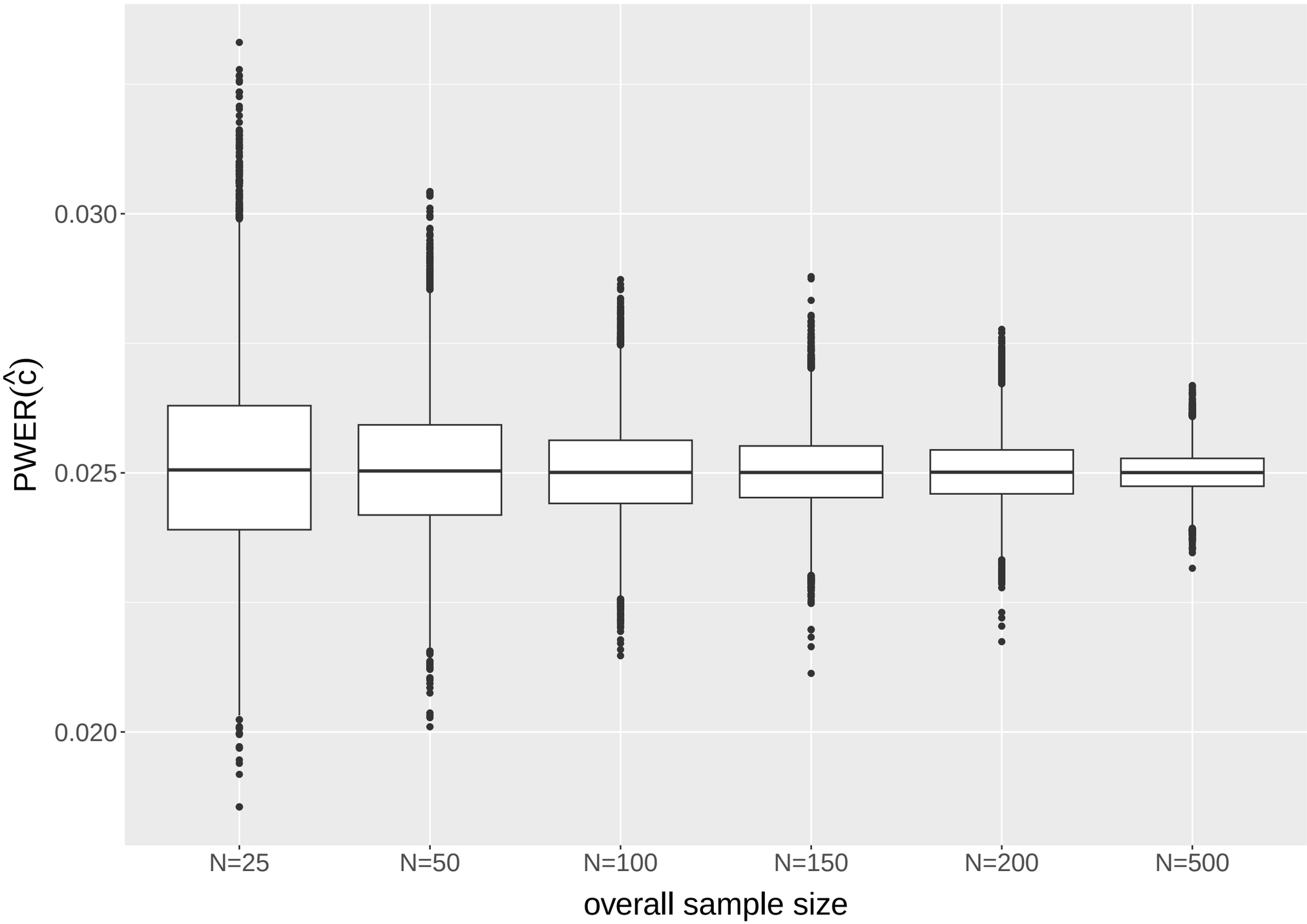
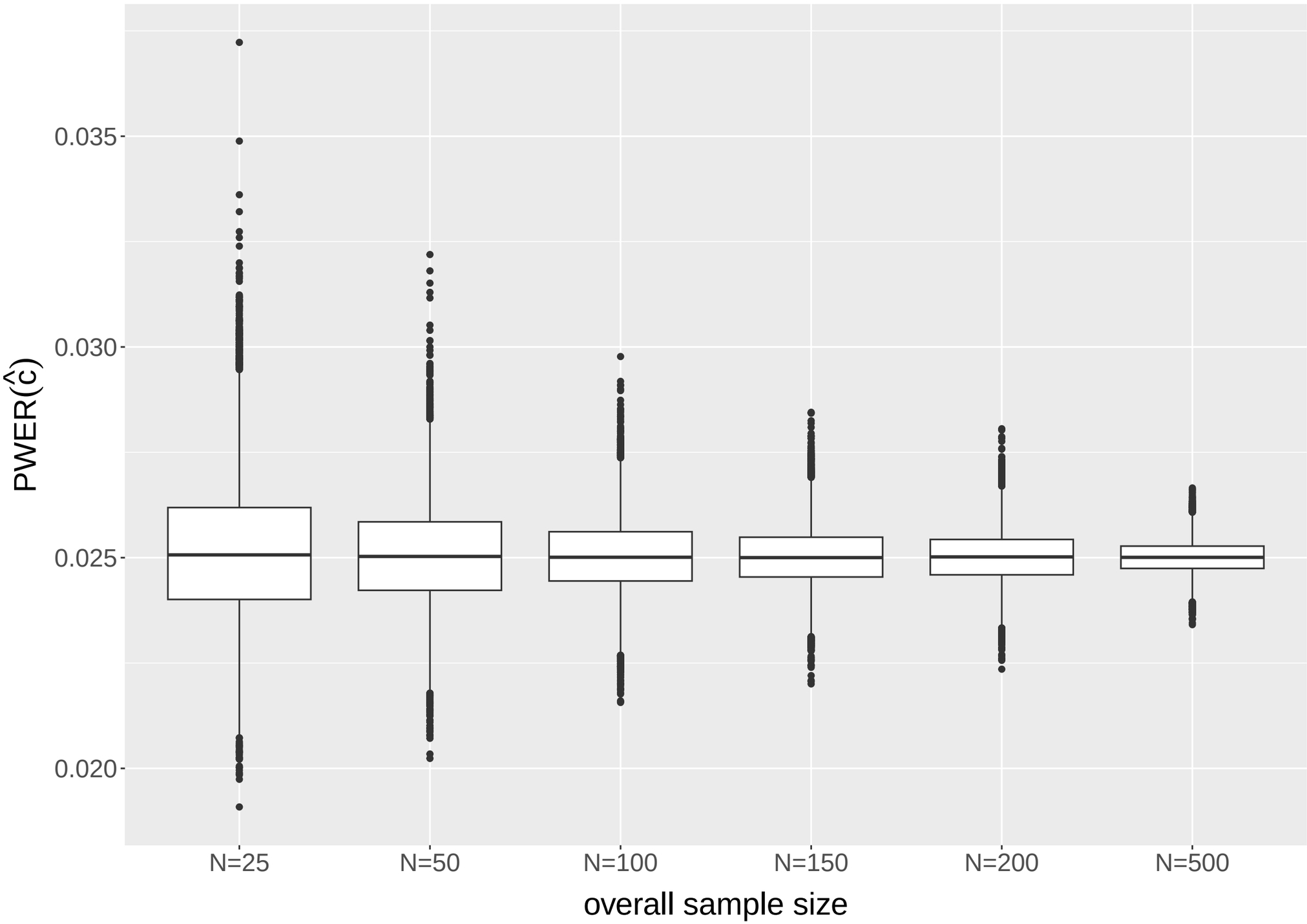

0.030
0.025
0.020
PWER($\hat{c}$)
N=25
N=50
N=100
N=150
N=200
N=500
overall sample size

Figure 10


0.035
0.030
0.025
0.020
PWER($\hat{c}$)
N=25
N=50
N=100
N=150
N=200
N=500
overall sample size

Distribution of the maximal strata-wise FWER

Figure	Prevalence estimate				Test statistics	Allocation in strata	sample size	number of biomarkers	Mean	SD	Min	Q1	Med	Q3	Max
11	MLE	independent biomarkers	random prevalences	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 2$	0.03922	0.006	0.02512	0.03461	0.03961	0.04422	0.04936
								$m = 3$	0.04597	0.0096	0.0256	0.03859	0.04519	0.05262	0.0731
								$m = 4$	0.04894	0.01125	0.02641	0.04036	0.04739	0.05592	0.09543
								$m = 5$	0.0502	0.01192	0.02722	0.04133	0.04822	0.05706	0.10456
								$m = 6$	0.05068	0.01196	0.02708	0.04203	0.04841	0.05713	0.11328
								$m = 7$	0.05079	0.0116	0.02842	0.04243	0.04868	0.05669	0.11746
								$m = 8$	0.05066	0.011	0.02895	0.04287	0.04865	0.05621	0.11428
12	MLE	correlated biomarkers	random prevalences	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 2$	0.03908	0.00595	0.02502	0.03486	0.03941	0.04386	0.04936
								$m = 3$	0.04591	0.00981	0.02505	0.03877	0.04476	0.05224	0.07311
								$m = 4$	0.04904	0.0117	0.02585	0.04048	0.0473	0.05565	0.09498
								$m = 5$	0.05027	0.01225	0.02685	0.04161	0.04807	0.05663	0.11552
								$m = 6$	0.05067	0.01205	0.0269	0.04236	0.0486	0.05659	0.13595
								$m = 7$	0.05072	0.01161	0.02669	0.04263	0.04867	0.05623	0.12456
								$m = 8$	0.05064	0.01128	0.026	0.04295	0.04863	0.05601	0.13021
13	MLE		fixed prevalences (equal in all strata)	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 2$	0.037	0.00059	0.0348	0.03659	0.03697	0.03742	0.03901
								$m = 3$	0.04277	0.00076	0.04005	0.04225	0.04275	0.04328	0.04579
								$m = 4$	0.04549	0.00083	0.04237	0.04493	0.04548	0.04605	0.04883
								$m = 5$	0.04677	0.00084	0.04395	0.0462	0.04676	0.04734	0.05026
								$m = 6$	0.04732	0.00081	0.04474	0.04677	0.0473	0.04786	0.05094
								$m = 7$	0.04752	0.00077	0.04477	0.04699	0.04751	0.04803	0.05062
								$m = 8$	0.04752	0.00073	0.04452	0.04702	0.04752	0.048	0.05069
14	MLE		fixed prevalences (0.5 in one stratum and equal in the others)	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 2$	0.03945	0.00061	0.03701	0.03906	0.03944	0.03988	0.04185
								$m = 3$	0.05156	0.00104	0.04779	0.05086	0.05155	0.05225	0.0562
								$m = 4$	0.05993	0.0014	0.05448	0.05898	0.05991	0.06086	0.0655
								$m = 5$	0.06568	0.00169	0.05902	0.06453	0.06566	0.06678	0.07392
								$m = 6$	0.06967	0.00194	0.06268	0.06835	0.06962	0.07094	0.07829
								$m = 7$	0.07246	0.00211	0.0652	0.07102	0.0724	0.07383	0.08103
								$m = 8$	0.07445	0.00226	0.06686	0.0729	0.07441	0.07593	0.08449
15	marginal prevalence estimator	independent biomarkers	random prevalences	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 2$	0.03923	0.00598	0.02512	0.03464	0.03958	0.04425	0.04936
								$m = 3$	0.04596	0.00958	0.0256	0.03861	0.04522	0.05261	0.0731
								$m = 4$	0.04893	0.01123	0.02641	0.04035	0.04741	0.05589	0.09429
								$m = 5$	0.0502	0.01193	0.02722	0.04133	0.0482	0.05708	0.10417
								$m = 6$	0.05068	0.01195	0.02708	0.04203	0.04843	0.05711	0.11248
								$m = 7$	0.05078	0.01159	0.02842	0.04243	0.04866	0.05671	0.11778
								$m = 8$	0.05066	0.011	0.02901	0.04287	0.04864	0.05623	0.11639
16	MLE	independent biomarkers	random prevalences	pairwise different treatments	z-distr.	equal	$N = 500$	$m = 2$	0.03924	0.00601	0.02512	0.03462	0.03962	0.04424	0.04937
								$m = 3$	0.04599	0.00961	0.02566	0.03862	0.0452	0.05266	0.07314
								$m = 4$	0.04897	0.01127	0.0265	0.04038	0.04744	0.05597	0.0957
								$m = 5$	0.05023	0.01194	0.02714	0.04136	0.04825	0.05711	0.10471
								$m = 6$	0.05073	0.01199	0.02707	0.04205	0.04845	0.05719	0.11345
								$m = 7$	0.05084	0.01163	0.02836	0.04246	0.04872	0.05675	0.1177
								$m = 8$	0.05075	0.01104	0.02898	0.04294	0.04871	0.05634	0.11489

Distribution of the maximal strata-wise FWER

Figure	Prevalence estimate				Test statistics	Allocation in strata	sample size	number of biomarkers	Mean	SD	Min	Q1	Med	Q3	Max
17	MLE	independent biomarkers	random prevalences	one single treatment	t-distr.	equal	$N = 500$	$m = 2$	0.03826	0.00629	0.02502	0.03323	0.03849	0.04349	0.04936
								$m = 3$	0.04466	0.00964	0.02534	0.03727	0.04387	0.05114	0.0731
								$m = 4$	0.04773	0.01104	0.02581	0.03951	0.04626	0.05453	0.09533
								$m = 5$	0.04933	0.01156	0.02598	0.04086	0.04757	0.05598	0.10323
								$m = 6$	0.05025	0.01154	0.02641	0.04194	0.04844	0.05653	0.11155
								$m = 7$	0.05087	0.01117	0.02767	0.04289	0.04918	0.05671	0.11511
								$m = 8$	0.05122	0.01064	0.02853	0.04375	0.04954	0.05675	0.11187
18	MLE	independent biomarkers	random prevalences	pairwise different treatments	t-distr.	random allocation probabilities	$N = 500$	$m = 2$	0.0392	0.00598	0.02512	0.03462	0.03952	0.04413	0.04936
								$m = 3$	0.04594	0.00958	0.02568	0.03855	0.04519	0.05258	0.07309
								$m = 4$	0.04891	0.01123	0.0264	0.04036	0.04738	0.05586	0.09548
								$m = 5$	0.05018	0.01191	0.02711	0.04133	0.0482	0.05706	0.10502
								$m = 6$	0.05067	0.01196	0.02706	0.042	0.04838	0.05718	0.11356
								$m = 7$	0.05079	0.01159	0.02815	0.04246	0.04869	0.05666	0.11745
								$m = 8$	0.05067	0.01099	0.0291	0.0429	0.04866	0.05622	0.11473
19	MLE	independent biomarkers	random prevalences	pairwise different treatments	t-distr.	equal	$N = 25$	$m = 3$	0.04515	0.00949	0.02501	0.03792	0.04444	0.05171	0.07199
							$N = 50$		0.04577	0.00971	0.02526	0.03839	0.04503	0.05244	0.07267
							$N = 100$		0.04589	0.00967	0.02555	0.03855	0.04505	0.05252	0.07293
							$N = 150$		0.04593	0.00965	0.02547	0.0386	0.0451	0.05264	0.07301
							$N = 200$		0.04594	0.00963	0.0255	0.03861	0.04513	0.05258	0.07302
							$N = 500$		0.04597	0.0096	0.0256	0.03859	0.04519	0.05262	0.0731
20	MLE	independent biomarkers	random prevalences	pairwise different treatments	t-distr.	equal	$N = 25$	$m = 4$	0.04663	0.00991	0.02621	0.03917	0.04561	0.05307	0.09162
							$N = 50$		0.04856	0.01121	0.02629	0.04015	0.04704	0.05545	0.09211
							$N = 100$		0.04884	0.01131	0.02635	0.04033	0.04722	0.05572	0.09342
							$N = 150$		0.04888	0.01127	0.02648	0.04037	0.04736	0.05582	0.09467
							$N = 200$		0.04892	0.01129	0.02636	0.04035	0.04741	0.05586	0.09398
							$N = 500$		0.04894	0.01125	0.02641	0.04036	0.04739	0.05592	0.09543

Figure 11

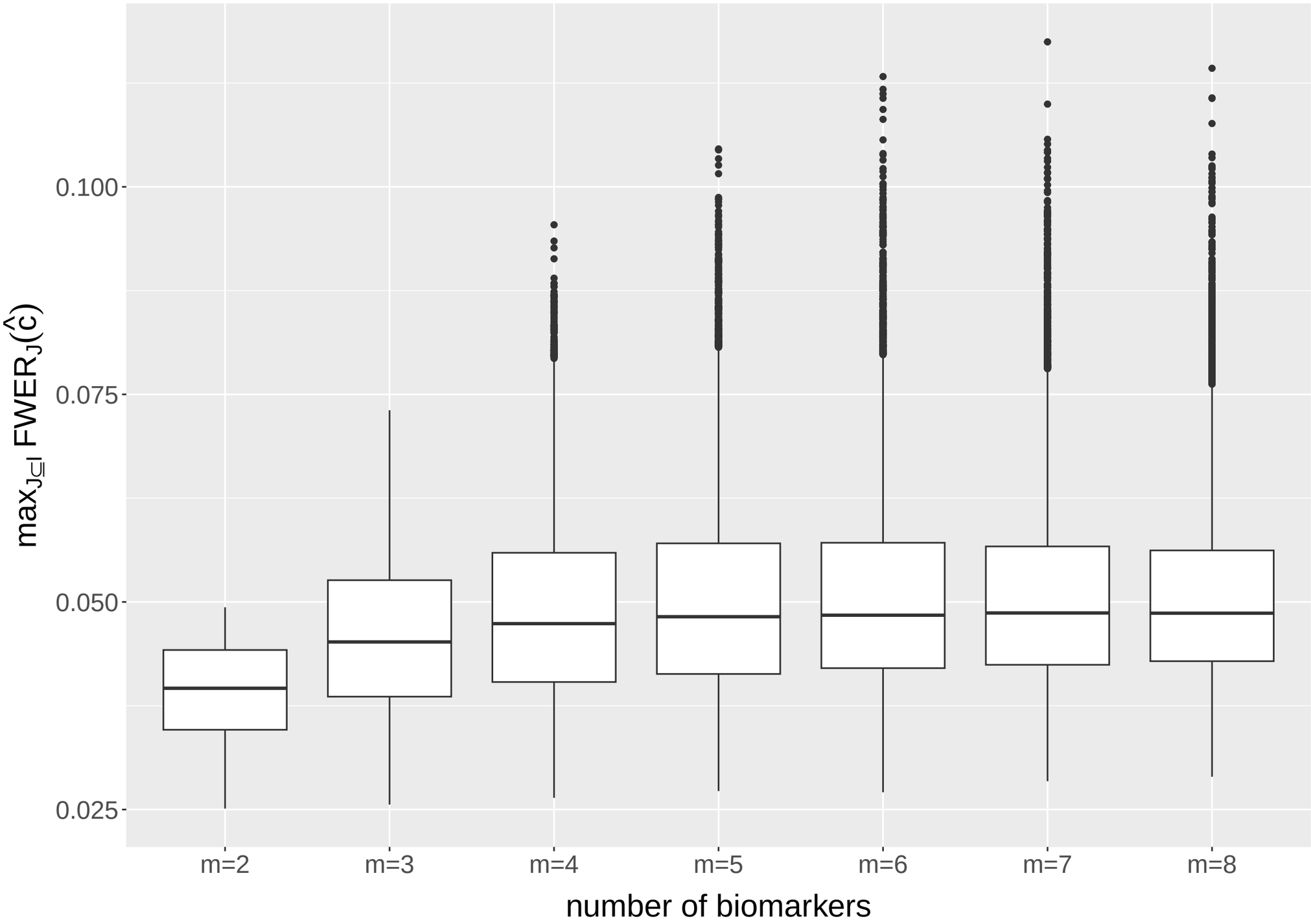

0.100
0.075
0.050
0.025
$\max_{J \subseteq I} FWER_J(\hat{c})$
m=2
m=3
m=4
m=5
m=6
m=7
m=8
number of biomarkers

Figure 12

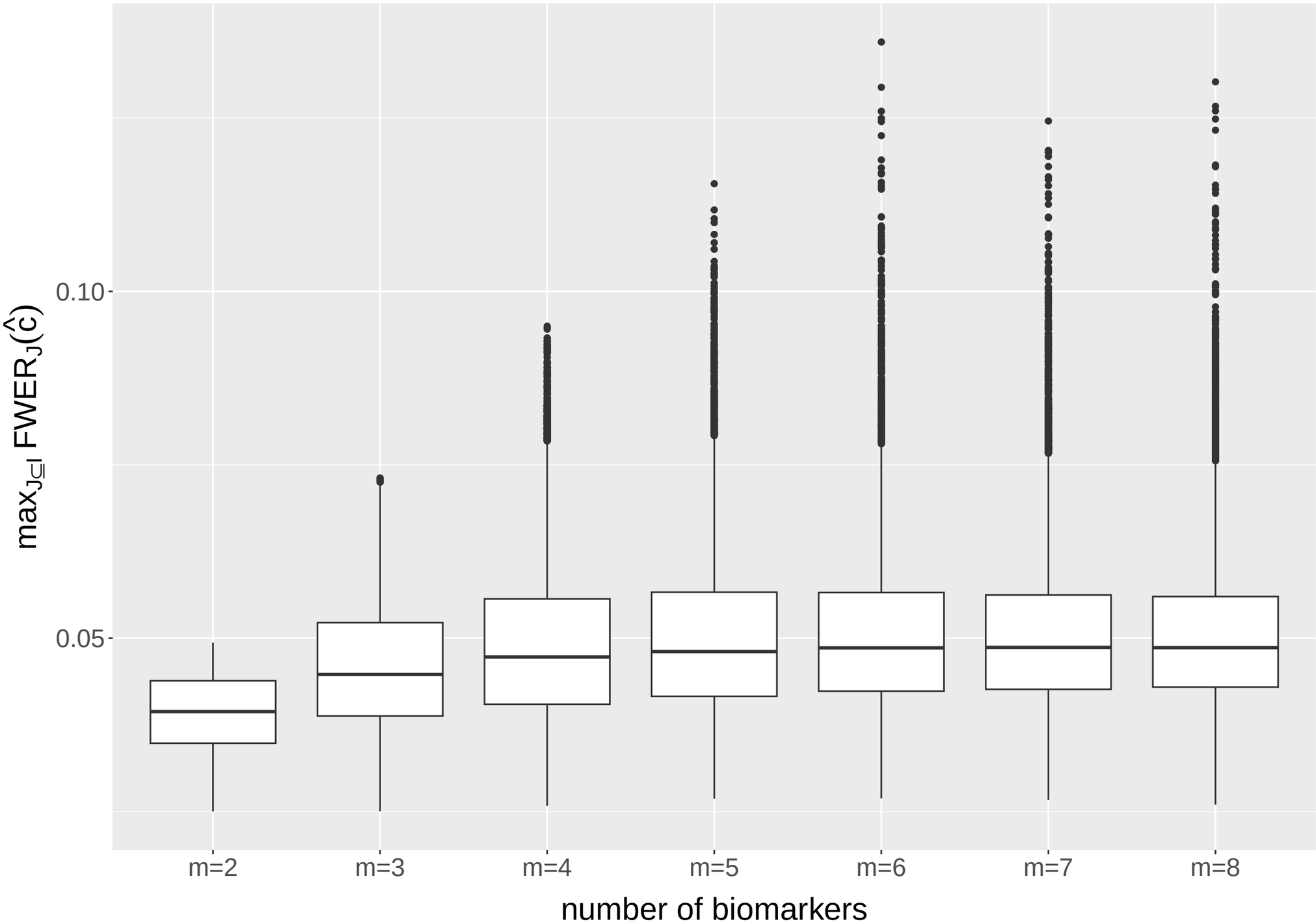

0.10
0.05
$\max_{J \subseteq I} \mathrm{FWER}_J(\hat{c})$
m=2
m=3
m=4
m=5
m=6
m=7
m=8
number of biomarkers

Figure 13

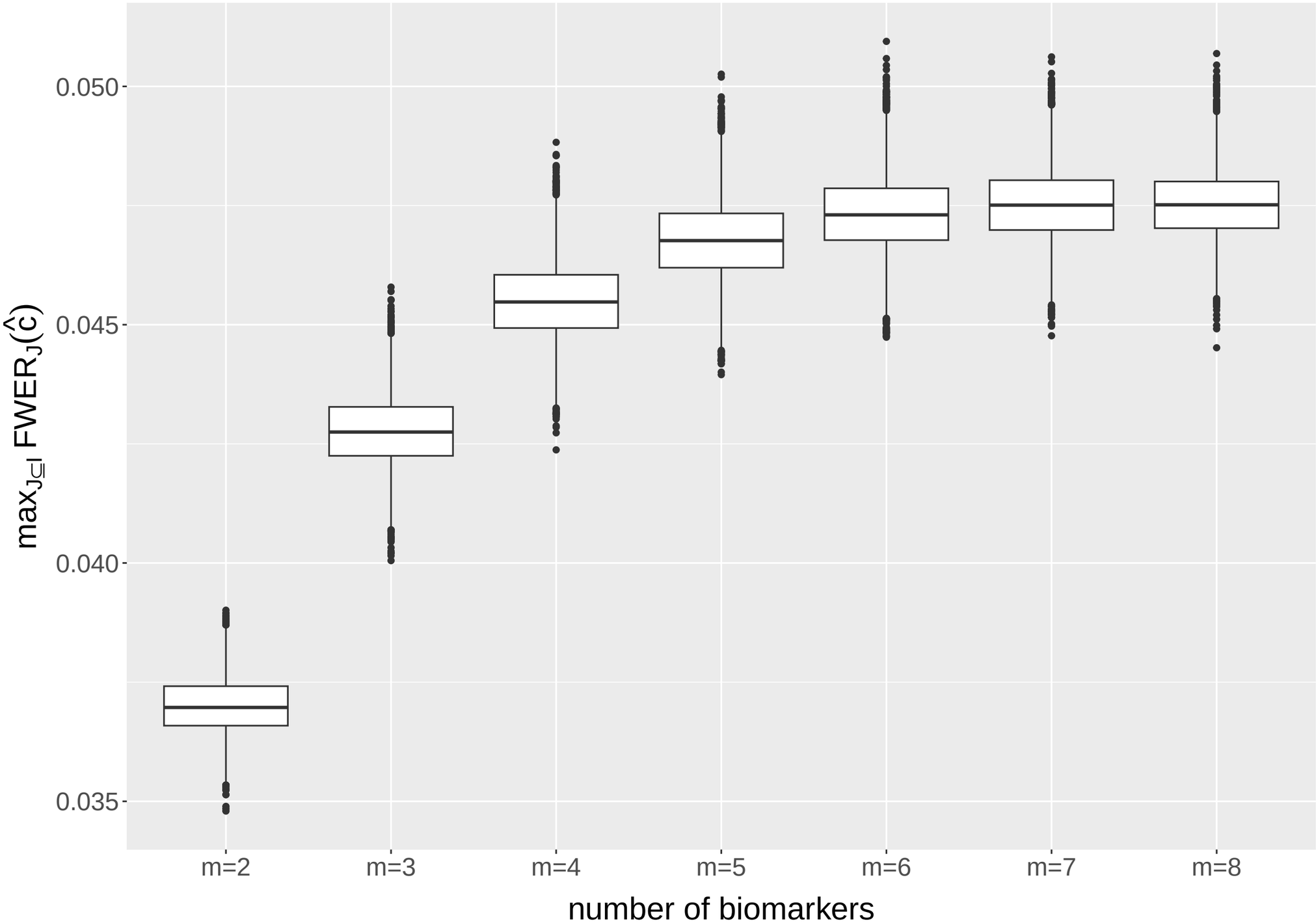

0.050
0.045
0.040
0.035
$\max_{J \subseteq I} \mathrm{FWER}_J(\hat{c})$
m=2
m=3
m=4
m=5
m=6
m=7
m=8
number of biomarkers

Figure 14

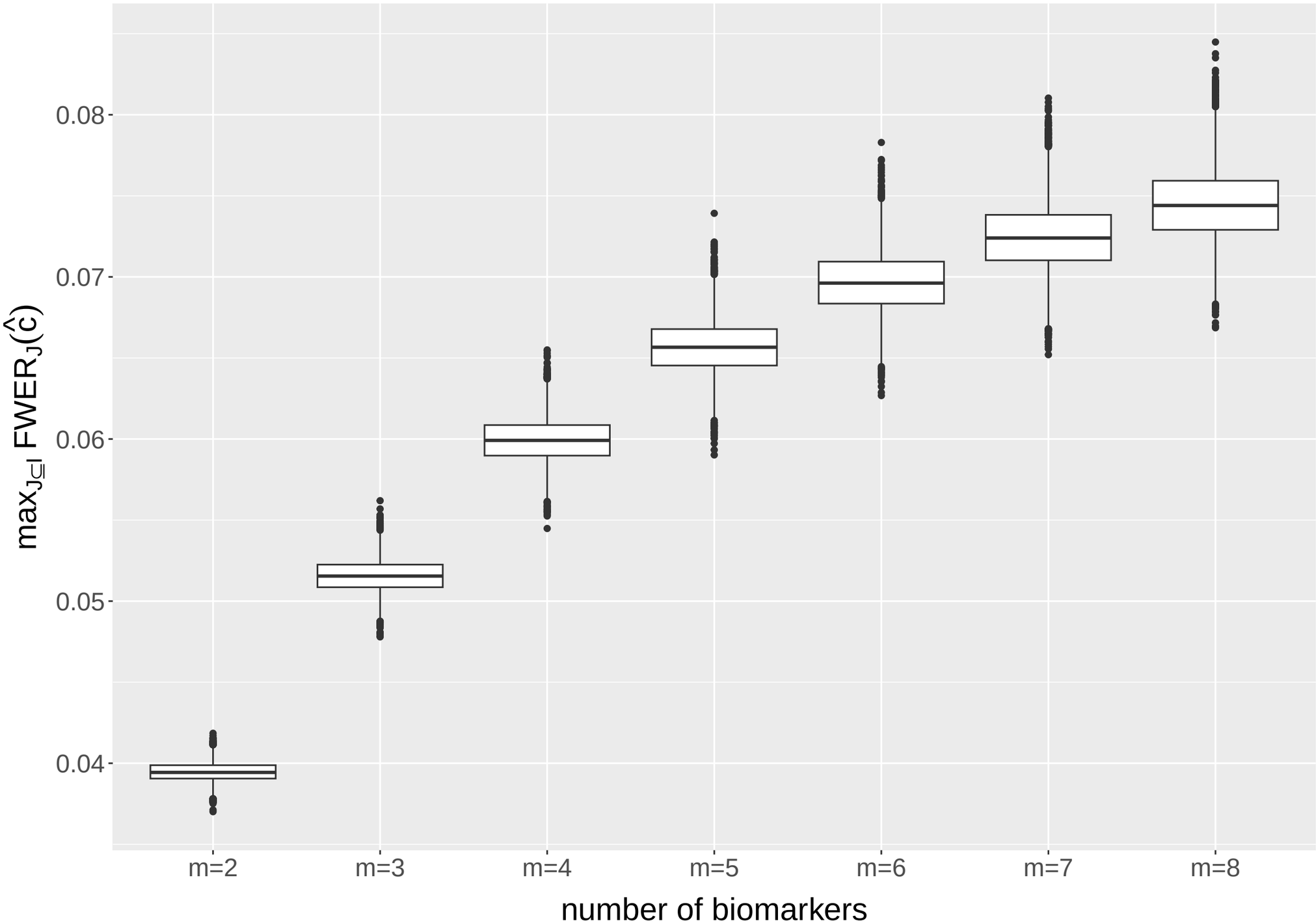

0.08
0.07
0.06
0.05
0.04
$\max_{J \subseteq I} FWER_J(\hat{c})$
m=2
m=3
m=4
m=5
m=6
m=7
m=8
number of biomarkers

Figure 15

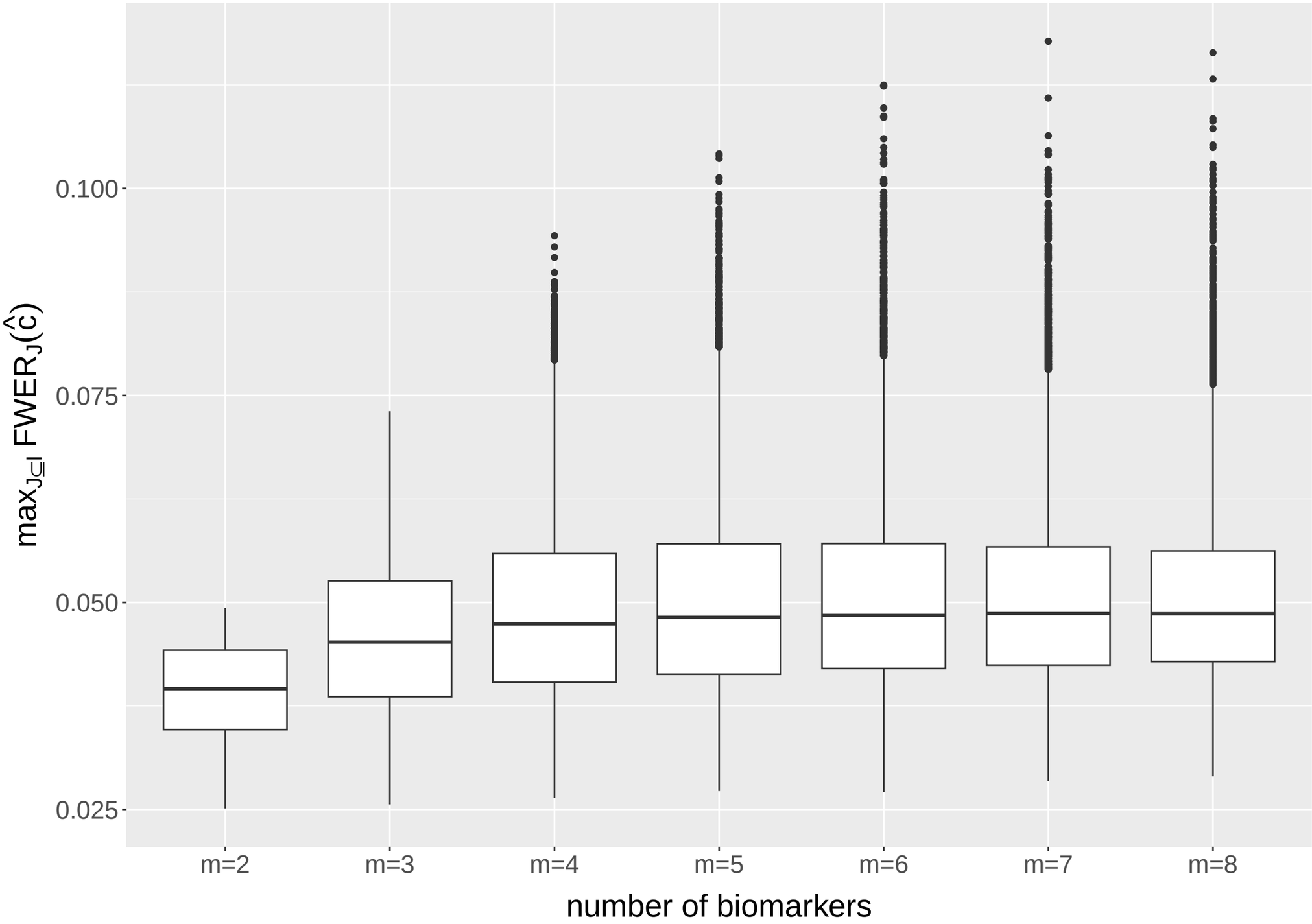

0.100
0.075
0.050
0.025
$\max_{J \subseteq I} FWER_J(\hat{c})$
m=2
m=3
m=4
m=5
m=6
m=7
m=8
number of biomarkers

Figure 16

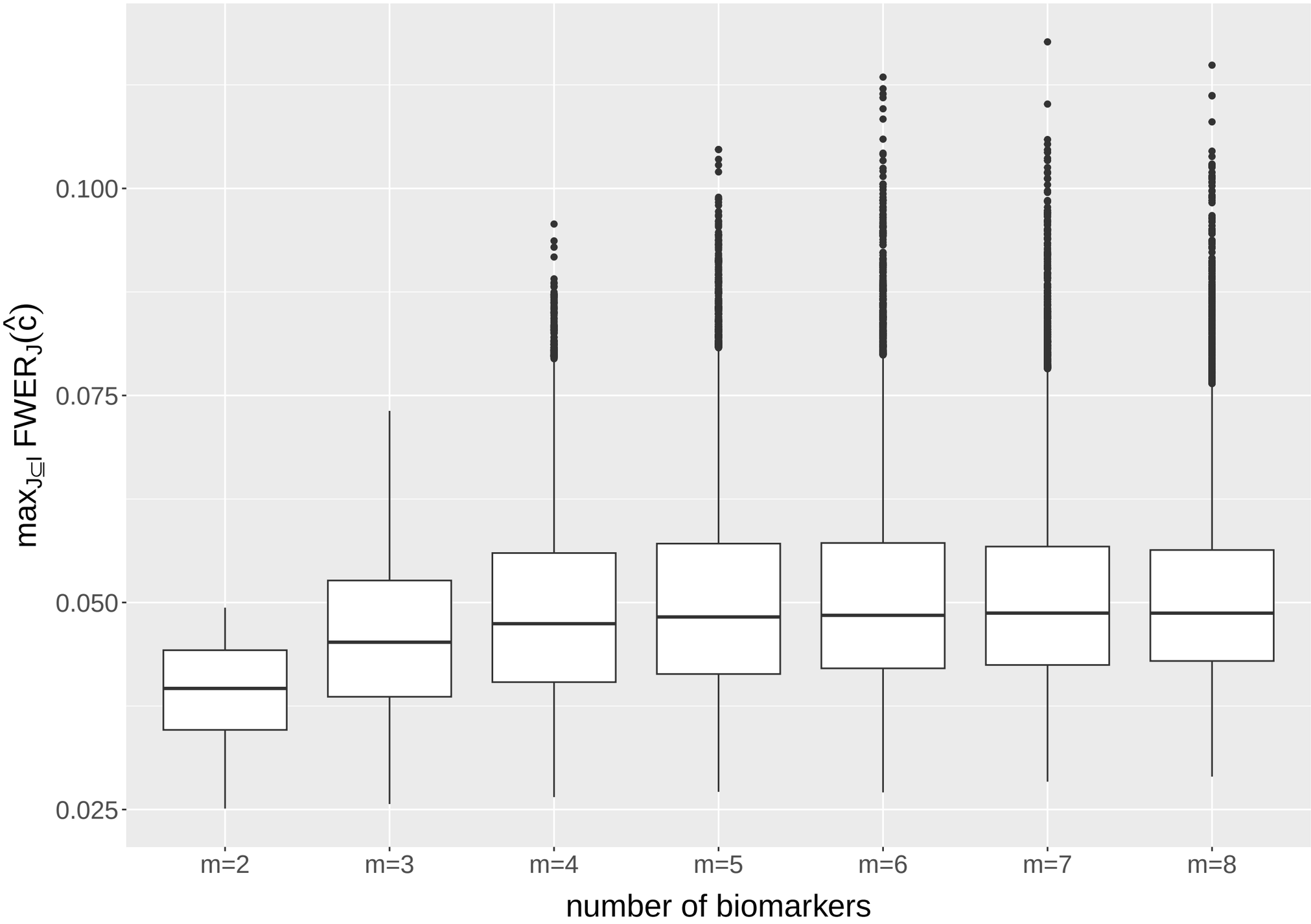

0.100
0.075
0.050
0.025
$\max_{J \subseteq I} \mathrm{FWER}_J(\hat{c})$
m=2
m=3
m=4
m=5
m=6
m=7
m=8
number of biomarkers

Figure 17

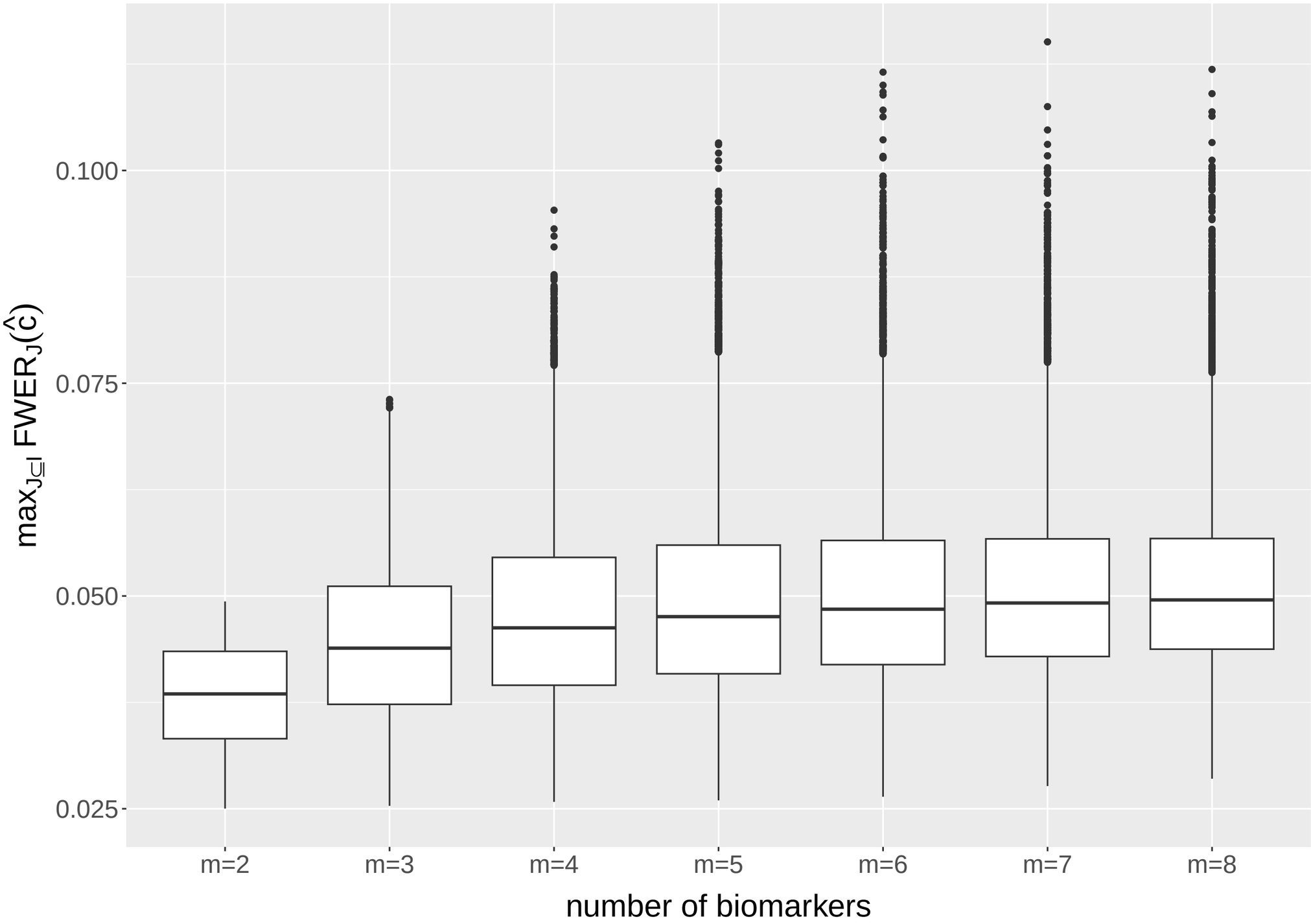

0.100
0.075
0.050
0.025
$\max_{J \subseteq I} \mathrm{FWER}_J(\hat{c})$
m=2
m=3
m=4
m=5
m=6
m=7
m=8
number of biomarkers

Figure 18

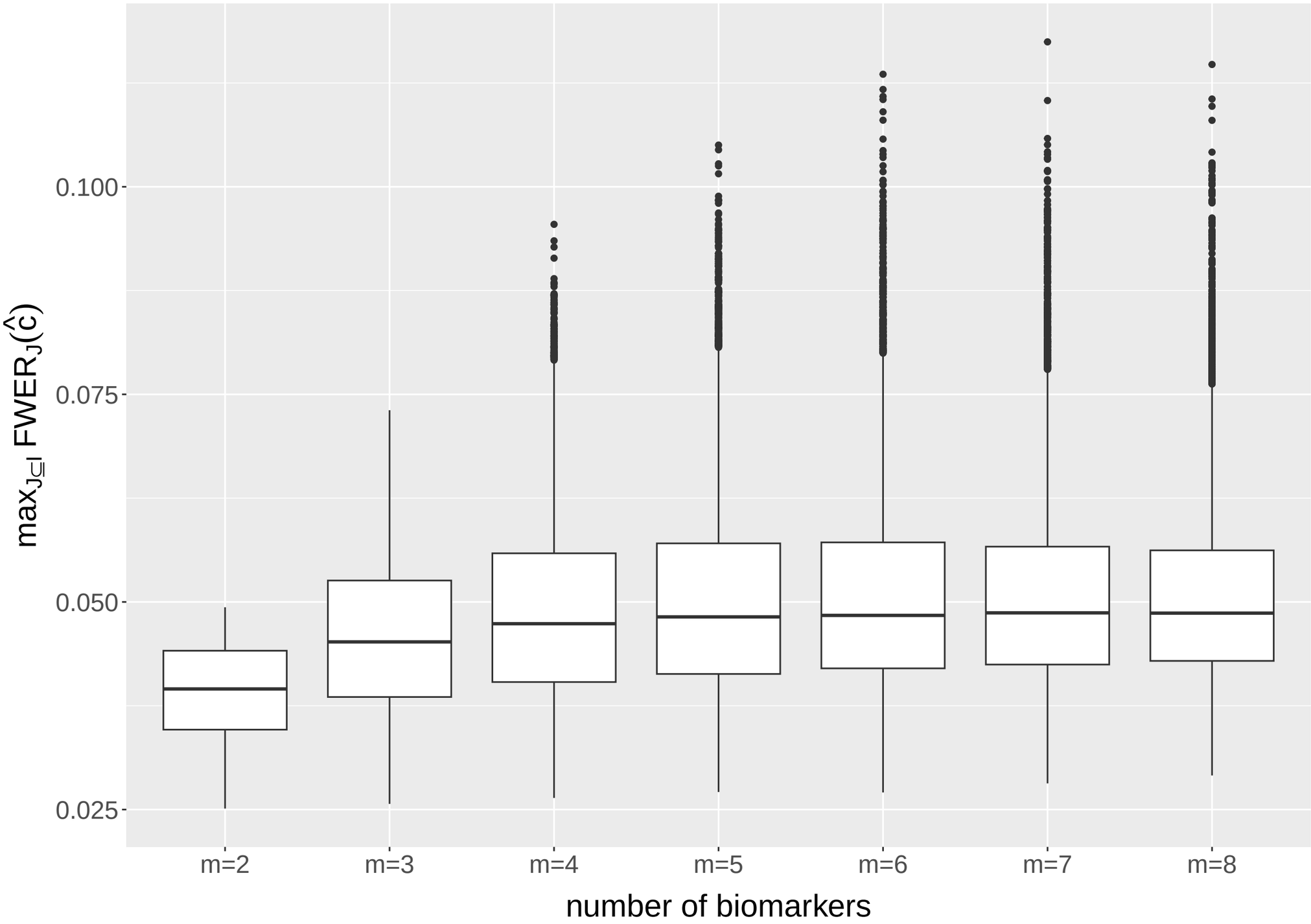

0.100
0.075
0.050
0.025
$\max_{J \subseteq I} FWER_J(\hat{c})$
m=2
m=3
m=4
m=5
m=6
m=7
m=8
number of biomarkers

Figure 19

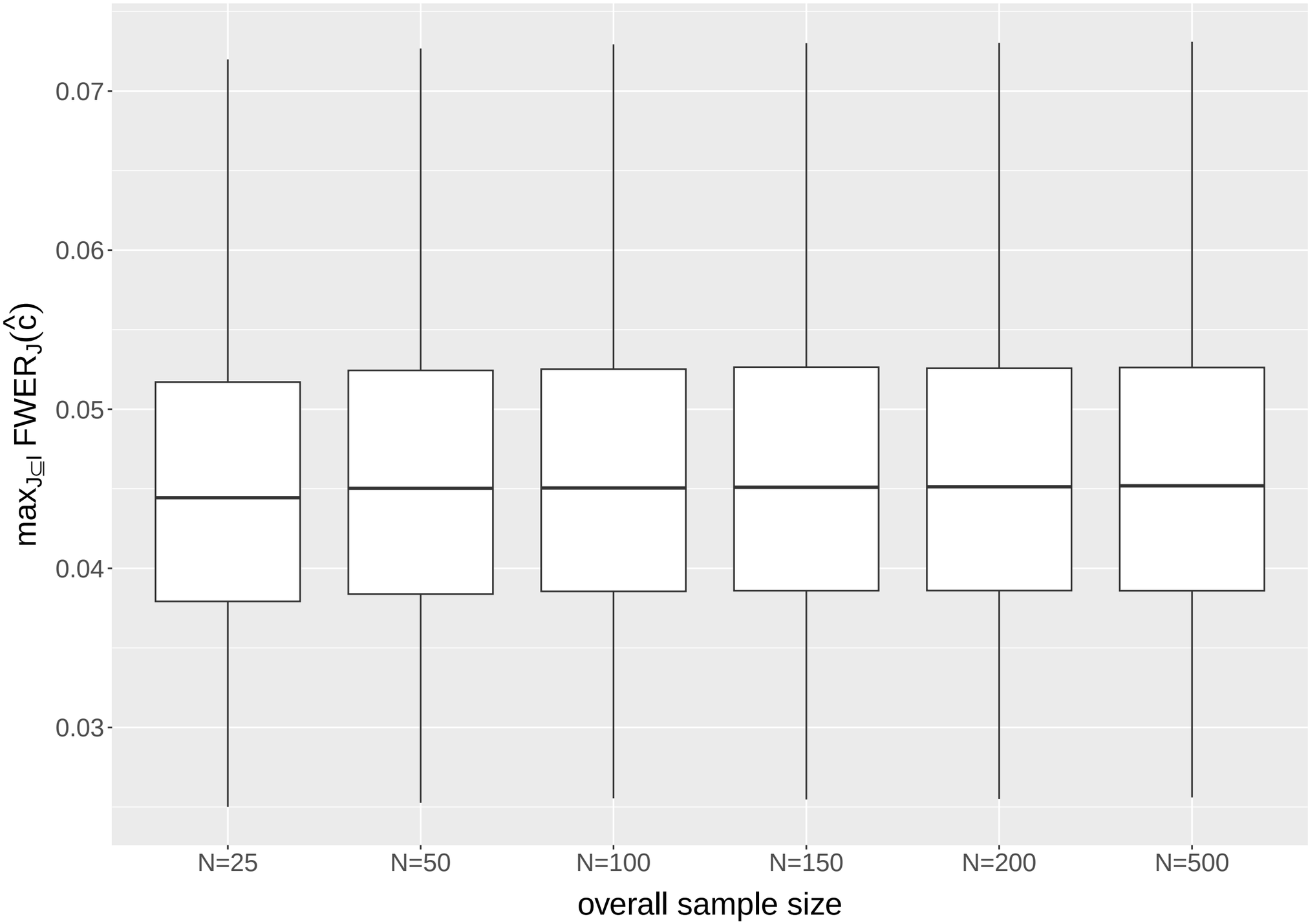
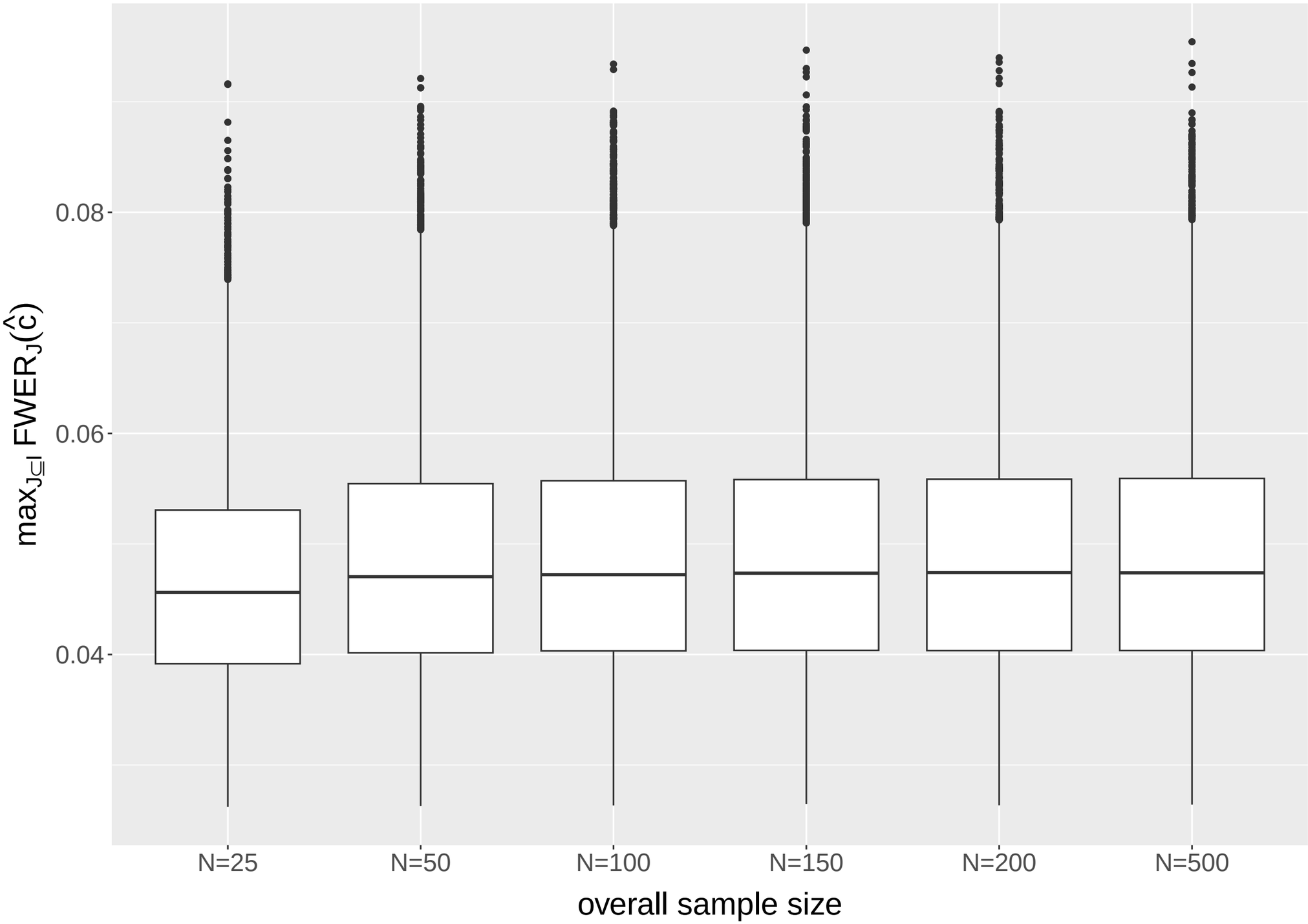

0.07
0.06
0.05
0.04
0.03
$\max_{J \subseteq I} FWER_J(\hat{c})$
N=25
N=50
N=100
N=150
N=200
N=500
overall sample size

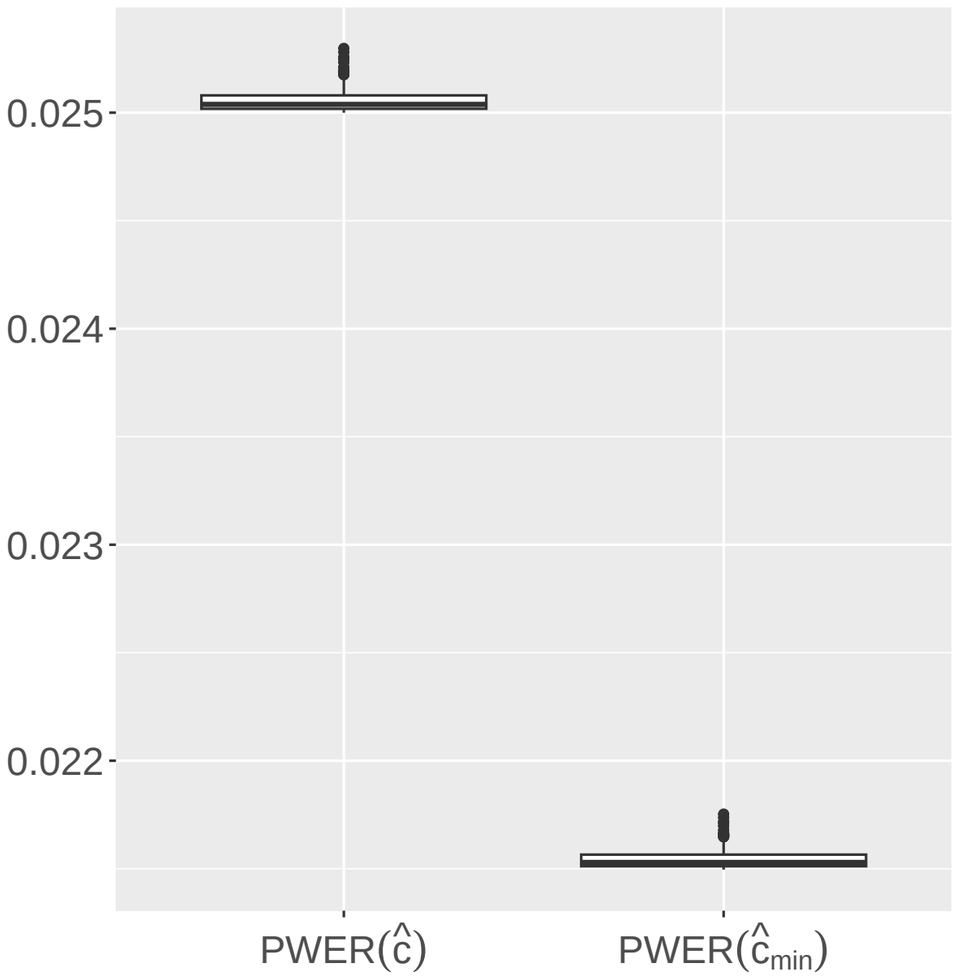
Figure 20


0.08
0.06
0.04
$\max_{J \subseteq I} FWER_J(\hat{c})$
N=25
N=50
N=100
N=150
N=200
N=500
overall sample size

Comparison of the true PWER and the maximal strata wise FWER, with and without the minimal prevalence adjustment

Figure	Prevalence estimate				Test statistics	Allocation in strata	sample size	number of biomarkers	Value	Mean	SD	Min	Q1	Med	Q3	Max
21	MLE	independent biomarkers (all with expression probability 0.1)	random prevalences	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 2$	$\mathrm{PWER}(\hat{c})$	0.02506	5e-05	0.025	0.02502	0.02504	0.02508	0.0253
									$\mathrm{PWER}(\hat{c}_{\min})$	0.02154	5e-05	0.0215	0.02151	0.02153	0.02157	0.02175
									$\max \mathrm{FWER}_J(\hat{c})$	0.04936	0	0.04936	0.04936	0.04936	0.04936	0.04936
									$\max \mathrm{FWER}_J(\hat{c}_{\min})$	0.04252	0	0.04252	0.04252	0.04252	0.04252	0.04252
22	MLE	independent biomarkers (all with expression probability 0.1)	random prevalences	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 3$	$\mathrm{PWER}(\hat{c})$	0.02502	2e-04	0.02412	0.0249	0.02502	0.02516	0.02584
									$\mathrm{PWER}(\hat{c}_{\min})$	0.01927	0.00033	0.01854	0.01904	0.01925	0.01949	0.0204
									$\max \mathrm{FWER}_J(\hat{c})$	0.07037	0.00128	0.06506	0.0695	0.07044	0.07127	0.07311
									$\max \mathrm{FWER}_J(\hat{c}_{\min})$	0.05449	3e-05	0.05437	0.05447	0.05449	0.05451	0.05525
23	MLE	independent biomarkers (all with expression probability 0.1)	random prevalences	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 4$	$\mathrm{PWER}(\hat{c})$	0.02501	0.00027	0.02388	0.02483	0.02501	0.02518	0.02611
									$\mathrm{PWER}(\hat{c}_{\min})$	0.01725	0.00041	0.01612	0.01696	0.01724	0.01754	0.0186
									$\max \mathrm{FWER}_J(\hat{c})$	0.08998	0.00229	0.08162	0.08842	0.09005	0.09159	0.09625
									$\max \mathrm{FWER}_J(\hat{c}_{\min})$	0.06274	0.00012	0.06219	0.06267	0.06271	0.06274	0.06346
24	MLE	independent biomarkers (all with expression probability 0.1)	random prevalences	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 5$	$\mathrm{PWER}(\hat{c})$	0.025	0.00031	0.02387	0.0248	0.025	0.02521	0.02625
									$\mathrm{PWER}(\hat{c}_{\min})$	0.0155	0.00043	0.0143	0.01519	0.01549	0.0158	0.01699
									$\max \mathrm{FWER}_J(\hat{c})$	0.10854	0.00319	0.09754	0.10637	0.10865	0.11082	0.11742
									$\max \mathrm{FWER}_J(\hat{c}_{\min})$	0.06844	0.00018	0.06731	0.06836	0.06846	0.06856	0.06906
25	MLE	independent biomarkers (all with expression probability 0.1)	random prevalences	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 6$	$\mathrm{PWER}(\hat{c})$	0.025	0.00034	0.02343	0.02478	0.025	0.02524	0.02663
									$\mathrm{PWER}(\hat{c}_{\min})$	0.01403	0.00042	0.01283	0.01373	0.01402	0.01432	0.01553
									$\max \mathrm{FWER}_J(\hat{c})$	0.12579	0.00413	0.11145	0.12292	0.12581	0.12866	0.13942
									$\max \mathrm{FWER}_J(\hat{c}_{\min})$	0.0723	0.00034	0.07074	0.07208	0.07233	0.07255	0.07324
26	MLE	independent biomarkers (all with expression probability 0.1)	random prevalences	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 7$	$\mathrm{PWER}(\hat{c})$	0.025	0.00037	0.02339	0.02476	0.025	0.02525	0.02653
									$\mathrm{PWER}(\hat{c}_{\min})$	0.01278	0.00039	0.01157	0.01251	0.01277	0.01305	0.01417
									$\max \mathrm{FWER}_J(\hat{c})$	0.14179	0.00502	0.12382	0.1383	0.14178	0.14524	0.15768
									$\max \mathrm{FWER}_J(\hat{c}_{\min})$	0.07479	0.00043	0.07303	0.07452	0.07481	0.0751	0.07592
27	MLE	independent biomarkers (all with expression probability 0.1)	random prevalences	pairwise different treatments	t-distr.	equal	$N = 500$	$m = 8$	$\mathrm{PWER}(\hat{c})$	0.02501	0.00039	0.02358	0.02475	0.02501	0.02527	0.02651
									$\mathrm{PWER}(\hat{c}_{\min})$	0.01178	0.00039	0.01059	0.01152	0.01178	0.01204	0.01326
									$\max \mathrm{FWER}_J(\hat{c})$	0.15656	0.00591	0.1368	0.15253	0.15653	0.16058	0.17744
									$\max \mathrm{FWER}_J(\hat{c}_{\min})$	0.07671	0.00045	0.07488	0.07641	0.07672	0.07702	0.07816

Figure 21


0.025
0.024
0.023
0.022
PWER($\hat{c}$)
PWER($\hat{c}_{min}$)

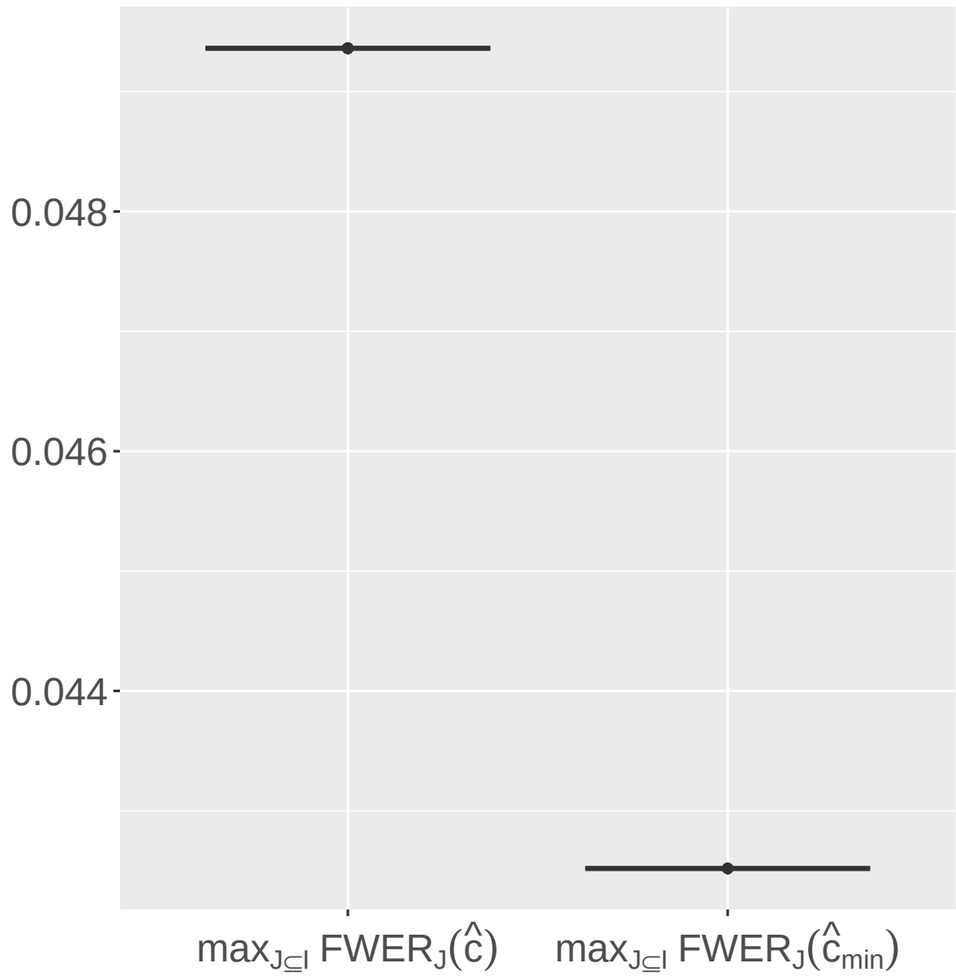
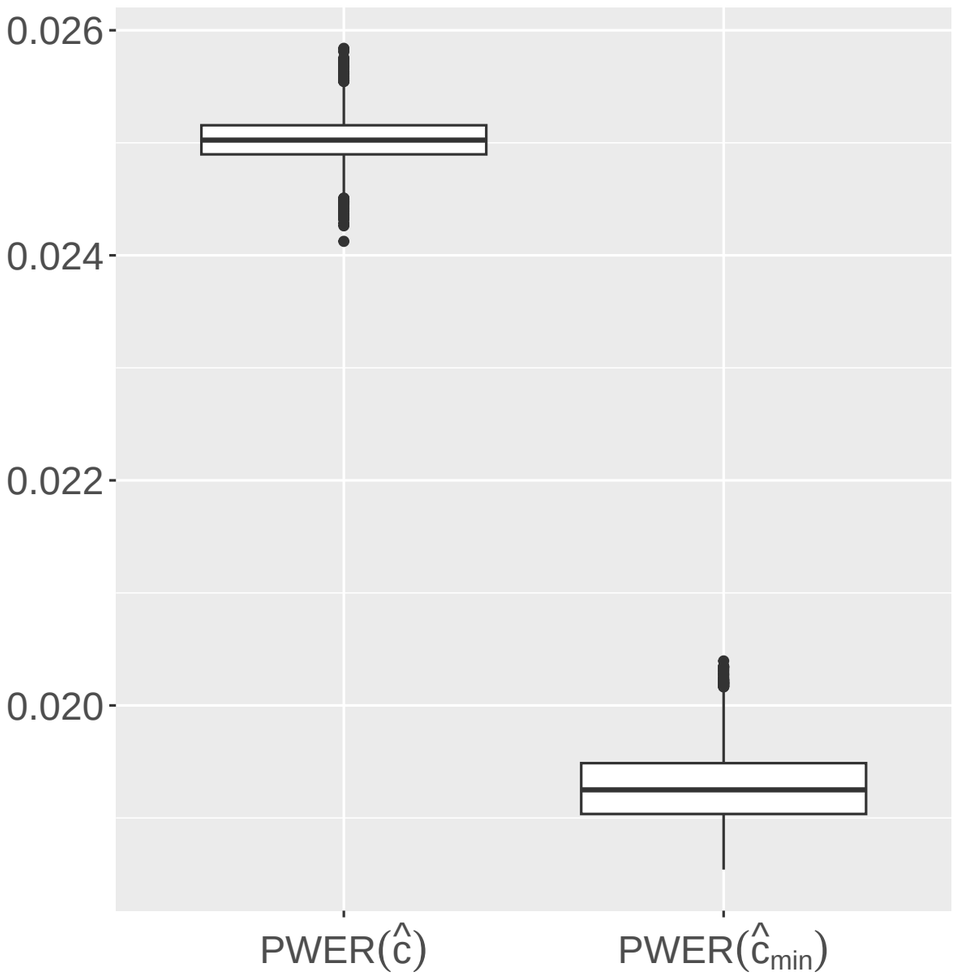

0.048
0.046
0.044
$\max_{J \subseteq I}$ FWER$_J$($\hat{c}$)
$\max_{J \subseteq I}$ FWER$_J$($\hat{c}_{min}$)

Figure 22


0.026
0.024
0.022
0.020
PWER($\hat{c}$)
PWER($\hat{c}_{min}$)

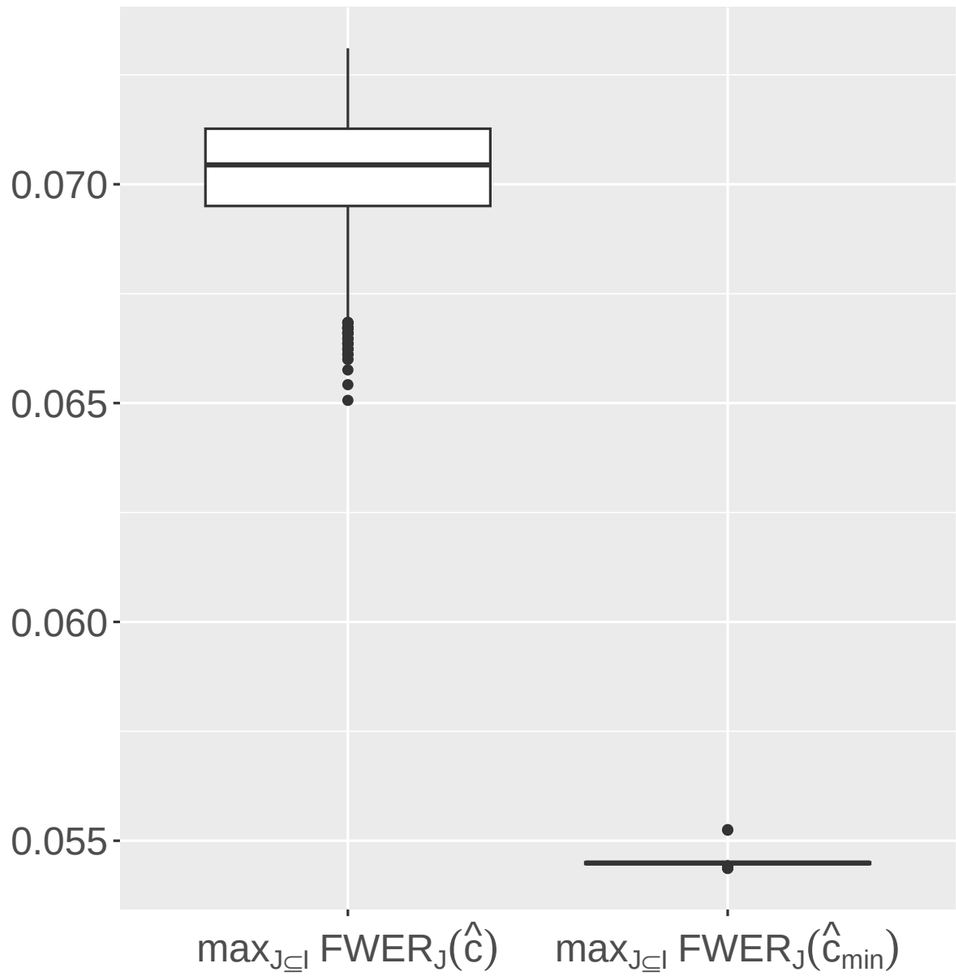
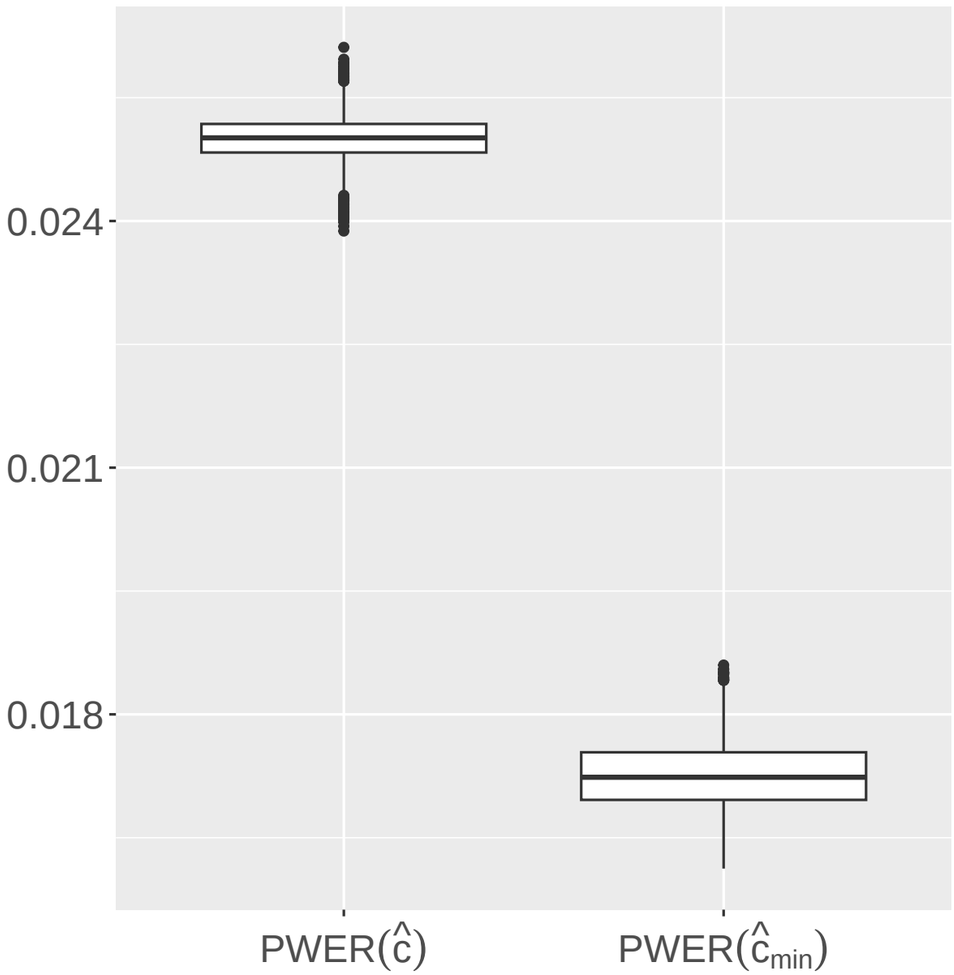

0.070
0.065
0.060
0.055
$\max_{J \subseteq I}$ FWER$_J$($\hat{c}$)
$\max_{J \subseteq I}$ FWER$_J$($\hat{c}_{min}$)

Figure 23


0.024
0.021
0.018
PWER($\hat{c}$)
PWER($\hat{c}_{min}$)

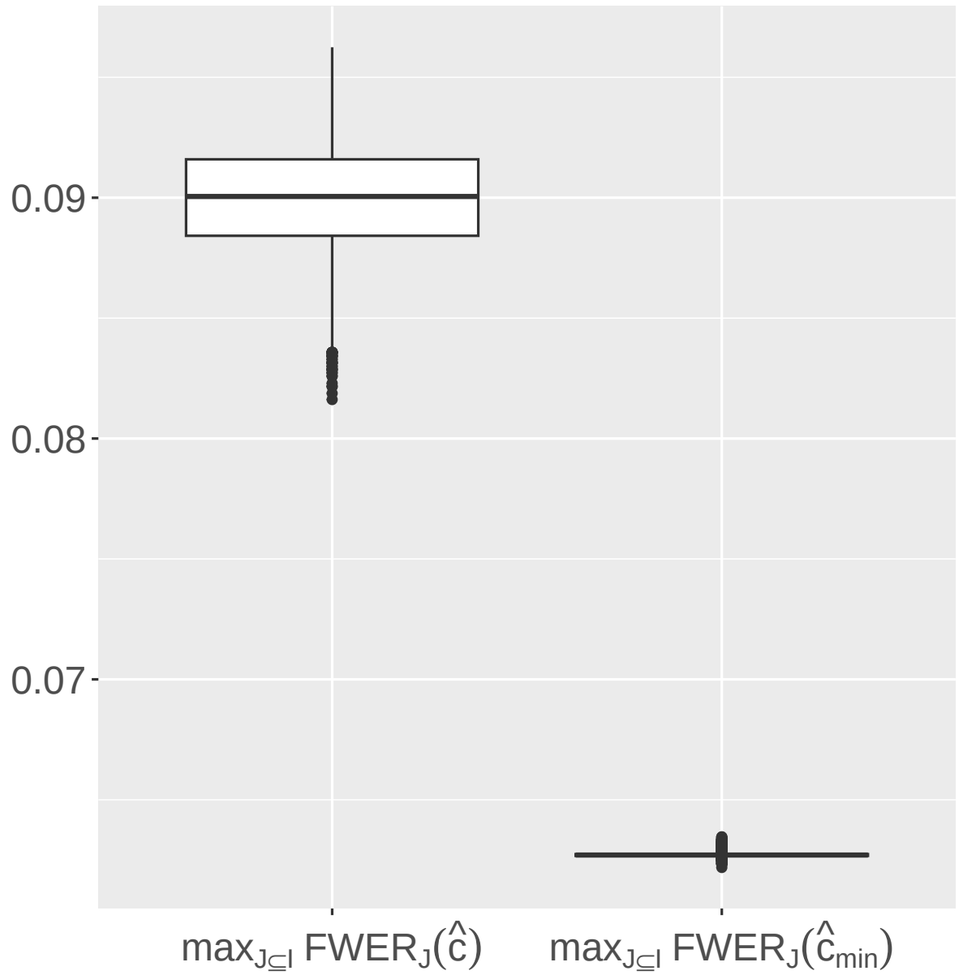

0.09
0.08
0.07
$\max_{J \subseteq I}$ FWER$_J$($\hat{c}$)
$\max_{J \subseteq I}$ FWER$_J$($\hat{c}_{min}$)

Figure 24

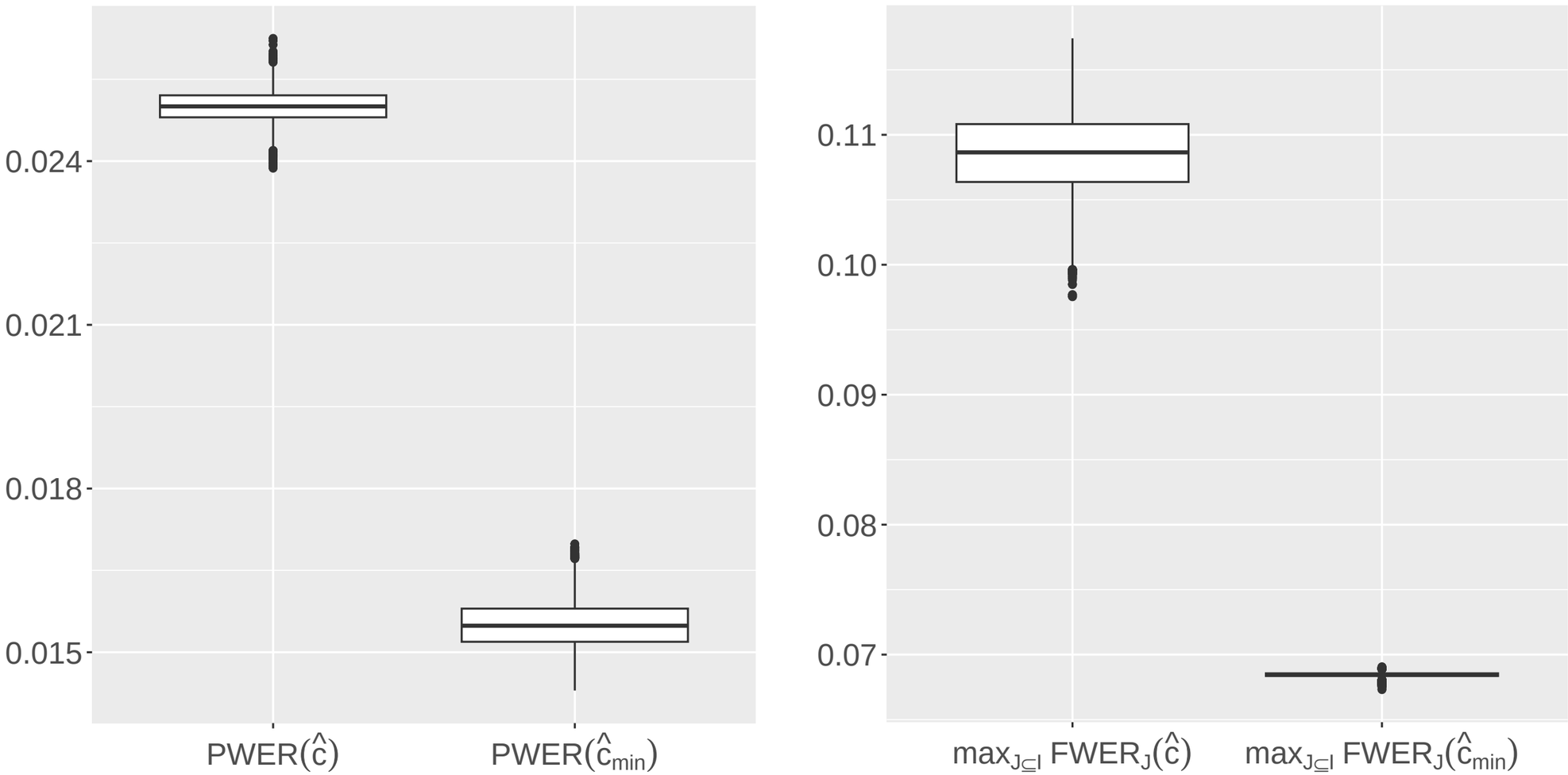

0.024
0.021
0.018
0.015
PWER($\hat{c}$)
PWER($\hat{c}_{min}$)
0.11
0.10
0.09
0.08
0.07
$\max_{J \subseteq I}$ FWER$_J$($\hat{c}$)
$\max_{J \subseteq I}$ FWER$_J$($\hat{c}_{min}$)

Figure 25

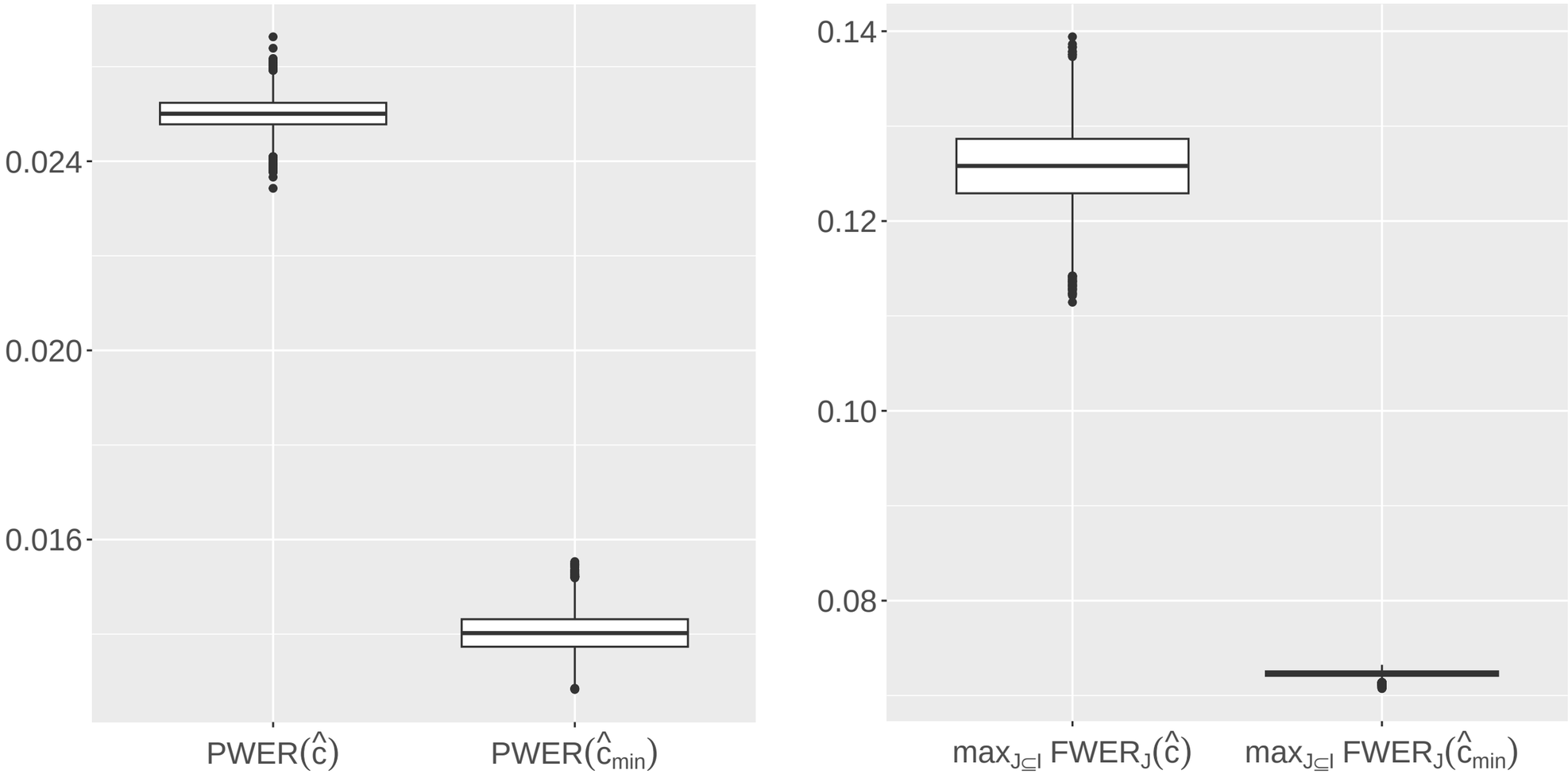

0.024
0.020
0.016
PWER($\hat{c}$)
PWER($\hat{c}_{min}$)
0.14
0.12
0.10
0.08
$\max_{J \subseteq I}$ FWER$_J$($\hat{c}$)
$\max_{J \subseteq I}$ FWER$_J$($\hat{c}_{min}$)

Figure 26

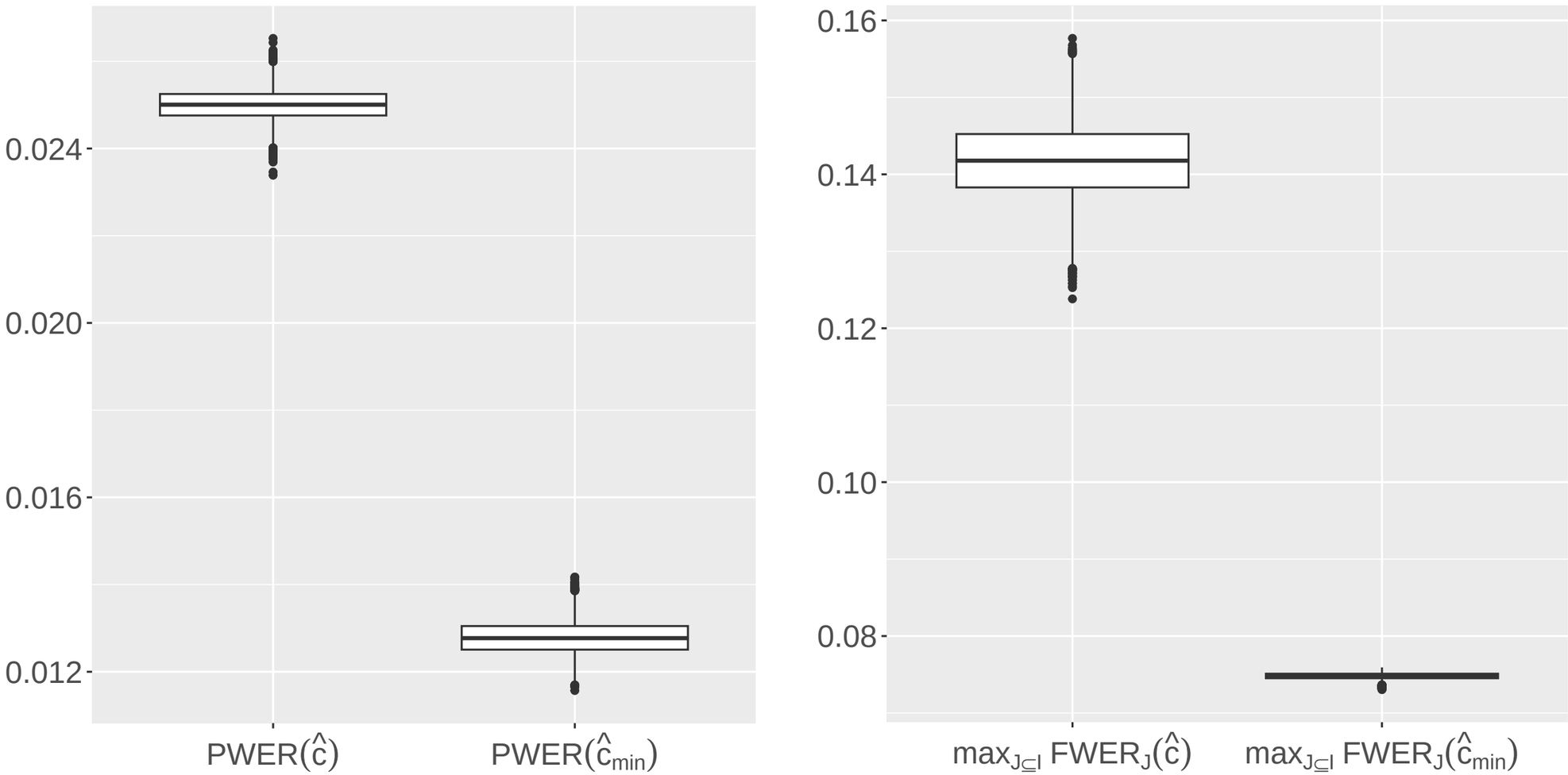
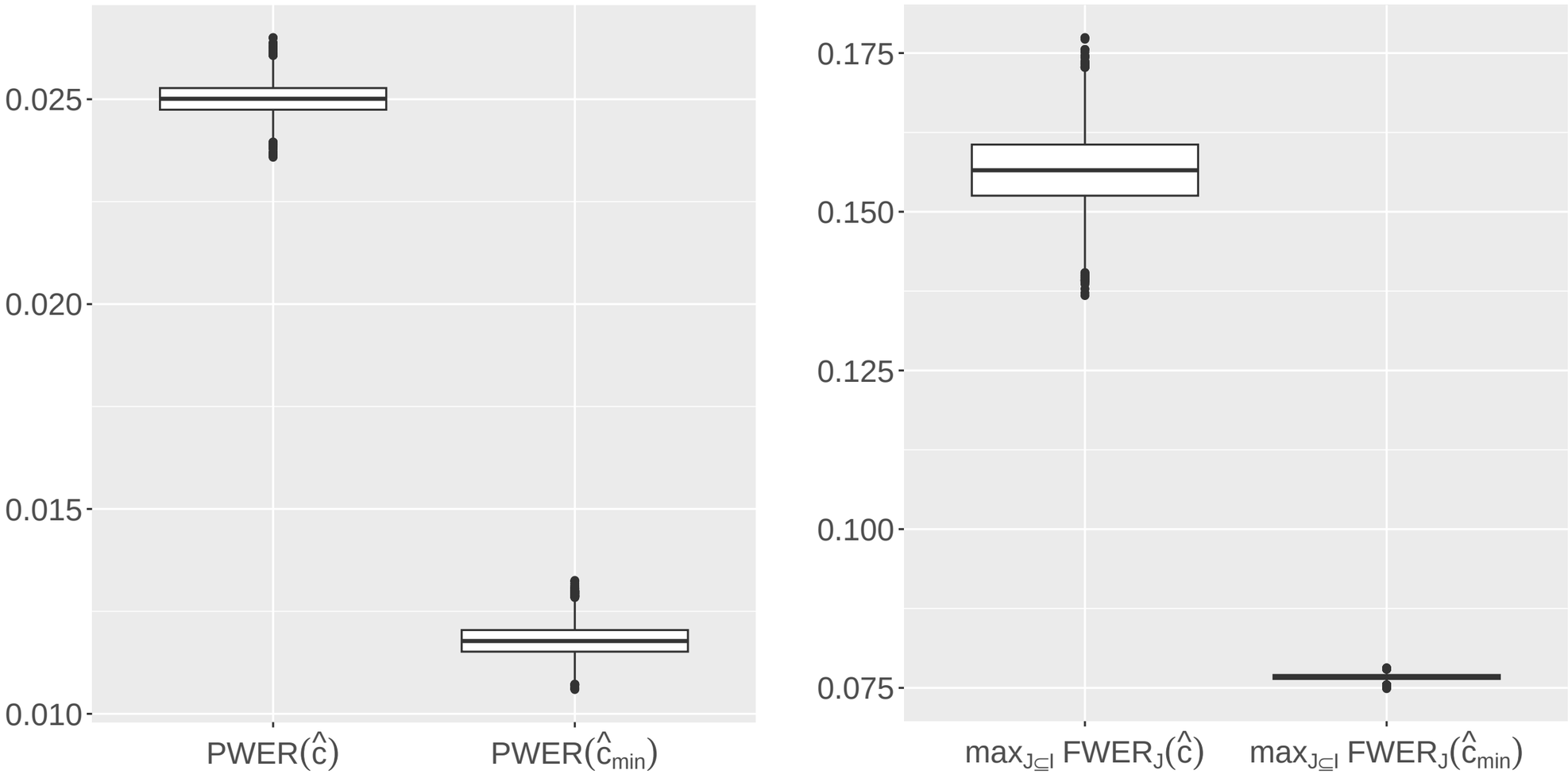

0.024
0.020
0.016
0.012
PWER($\hat{c}$)
PWER($\hat{c}_{min}$)
0.16
0.14
0.12
0.10
0.08
$\max_{J \subseteq I}$ FWER$_J$($\hat{c}$)
$\max_{J \subseteq I}$ FWER$_J$($\hat{c}_{min}$)

Figure 27


0.025
0.020
0.015
0.010
PWER($\hat{c}$)
PWER($\hat{c}_{min}$)
0.175
0.150
0.125
0.100
0.075
$\max_{J \subseteq I}$ FWER$_J$($\hat{c}$)
$\max_{J \subseteq I}$ FWER$_J$($\hat{c}_{min}$)